

PARALLEL PUMP STATION VER 1

Operator Training Simulator · cat-cracker feed pumps P-200A/B

Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University

HAZOP REPORT

Hazard and Operability Study of the cat-cracker feed pump station

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Teaching document generated from the HAZOP tab of the pump-station simulator. It is not a study of a real plant.

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HAZOP CLOSE-OUT SUMMARY

Project: Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator · Document PPS-V1-SA-HZP-001 rev A · 33 actions, all open at issue. The action party fills "Action taken" and the HAZOP leader changes the status to CLOSED when the close-out sheet (Appendix 3) is signed.

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N1 Tank-farm transfer and T200	R01	N1.1, N1.3	Give LSL-200 its own level transmitter LT-200B (today the LI-200 alarm, the LSL-200 start permissive and the LSL-200 trip read one measurement).	Instrument		OPEN	Risk 8 Medium
N1 Tank-farm transfer and T200	R02	N1.2, N1.4	Independent high-high level LSHH-200 on T200 that stops the tank-farm transfer (overfill protection to API 2350).	Instrument / Operations		OPEN	Risk 8 Medium
N1/N2 Tank-farm transfer and T200 / Suction	R03	N1.5, N2.3	Feed quality at the tank farm: vapour-pressure / flash-point check of each transfer and a line-up check, so naphtha or light material cannot reach T200 unseen.	Operations / Process		OPEN	Risk 8 Medium
N1/N7 Tank-farm transfer and T200 / D-150 and the cat-cracker interface	R04	N1.6, N7.3	T200 water draw-off routine (each shift) and water detection at the tank outlet before the feed is heated in E-100.	Operations		OPEN	Risk 8 Medium
N1 Tank-farm transfer and T200	R05	N1.7	Confirm the T200 pressure/vacuum vent (API 2000) for the largest pump-out and fill rates; inspect the flame arrester.	Mechanical		OPEN	Risk 6 Medium
N1/N3 Tank-farm transfer and T200 / Pump A leg	R06	N1.8, N3.8	Remotely operated, fire-safe emergency isolation valve EIV-200 on the T200 outlet, closable from the control room.	Process / Safety		OPEN	Risk 12 High
N2 Suction	R07	N2.1, N2.2, N2.6	Suction pressure transmitter PIT-209 with a low alarm in the DCS, and a differential-pressure gauge across S-200.	Instrument		OPEN	Risk 9 Medium
N2/N3 Suction / Pump A leg	R08	N1.5, N2.2, N2.3, N3.9	Vibration transmitters on P-200A/B with DCS alarms (VI-210/220 are local today).	Instrument / Mechanical		OPEN	Risk 9 Medium
N3/N7/N8 Pump A leg / D-150 and the cat-cracker interface / Procedures	R09	N3.1, N4.4, N5.2, N7.2, N8.4	Auto-start of the spare pump on loss of the running pump (motor trip or low header pressure), with the zero-speed permissive of R15.	Instrument / Operations		OPEN	Risk 12 High
N3/N6/N8 Pump A leg / Preheater E-100 / Procedures	R10	N3.1, N6.2, N8.1, N8.6	Steam cut-off: close TV-110 automatically when the feed flow is lost (FT-100 low or both pumps stopped).	Instrument		OPEN	Risk 12 High
N3/N5/N8 Pump A leg / Discharge header, FE-100, V100 and FRC-100 / Procedures	R11	N3.2, N4.5, N5.1, N8.3	Minimum-flow recycle from each pump discharge (upstream of the check valve) back to T200, sized for the API 610 minimum continuous flow.	Process / Mechanical		OPEN	Risk 12 High
N3/N5/N7 Pump A leg / Discharge header, FE-100, V100 and FRC-100 / D-150 and the cat-cracker interface	R12	N3.2, N4.5, N5.1, N7.1	Low-flow trip of the running pump (flow in each pump leg, or FT-100 low with V100 closed, time-delayed) so a blocked discharge stops the motor before the casing gets hot.	Instrument		OPEN	Risk 12 High
N3/N4 Pump A leg / Pump B leg	R13	N2.5, N3.3	Pump condition monitoring: performance test against the curve every quarter, wear-ring clearances at overhaul.	Mechanical		OPEN	Risk 8 Medium
N3/N8 Pump A leg / Procedures	R14	N3.4, N8.11	Rotation (bump) test and phase-sequence check in the close-out of any electrical work on the pump motors.	Electrical		OPEN	Risk 9 Medium

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N4/N8 Pump B leg / Procedures	R15	N4.2, N8.7	Zero-speed / reverse-rotation switch as a start permissive for each pump (or a non-reverse ratchet on the coupling).	Electrical / Mechanical		OPEN	Risk 9 Medium
N3/N4/N8 Pump A leg / Pump B leg / Procedures	R16	N2.1, N2.4, N3.6, N4.4, N8.2, N8.10	Dry-run protection: under-power relay on each motor, and a vent sight glass to prove the casing is liquid-full before a start.	Electrical / Operations		OPEN	Risk 12 High
N3/N4 Pump A leg / Pump B leg	R17	N3.8, N8.9	Dual mechanical seals (API 682) with seal-leak detection; seal drip pans with a kerb to the oily-water sewer.	Mechanical		OPEN	Risk 12 High
N3/N4 Pump A leg / Pump B leg	R18	N3.9, N4.1	Casing pressure transmitters PIT-210/220 with a high alarm on the idle pump (Rev C option on the SIS tab).	Instrument		OPEN	Risk 12 High
N4/N8 Pump B leg / Procedures	R19	N4.1, N4.2, N8.7	MOV discharge valves V201/V205 interlocked with the motors (Rev C option): open after the start, close on every stop.	Instrument / Electrical		OPEN	Risk 12 High
N4 Pump B leg	R20	N4.1, N5.5	Check valves: non-slam type in an inspection and seat-test programme; consider a second check valve in series on each leg.	Mechanical		OPEN	Risk 12 High
N4/N5/N8 Pump B leg / Discharge header, FE-100, V100 and FRC-100 / Procedures	R21	N2.7, N4.3, N4.6, N5.8, N8.6, N8.8	Thermal relief valve on each blocked-in liquid section: pump casing / discharge between check valve and block valve, and the header between the check valves and V100.	Process / Mechanical		OPEN	Risk 12 High
N5 Discharge header, FE-100, V100 and FRC-100	R22	N5.5	Close V100 automatically (FRC-100 to MAN 0 %) when both pumps are stopped, so D-150 cannot drain back through a passing check valve.	Instrument		OPEN	Risk 8 Medium
N5 Discharge header, FE-100, V100 and FRC-100	R23	N5.6	Plug or lock the header low-point drain; keep a lock / car-seal register of drains and vents.	Operations		OPEN	Risk 9 Medium
N5 Discharge header, FE-100, V100 and FRC-100	R24	N3.2, N3.7	Header pressure transmitter PIT-100 upstream of FE-100/V100 with a high alarm (blocked outlet, pump at shut-off) and a low alarm (pump lost).	Instrument		OPEN	Risk 12 High
N5 Discharge header, FE-100, V100 and FRC-100	R25	N3.3, N5.2, N5.4	FRC-100 output-high alarm (V100 above 95 % = loop saturated) and a high clamp on the remote set point.	Instrument		OPEN	Risk 9 Medium
N5/N7 Discharge header, FE-100, V100 and FRC-100 / D-150 and the cat-cracker interface	R26	N5.3, N5.6, N7.2, N7.4, N7.6	Flow-balance alarm FT-100 against FI-151 and the D-150 level trend, to catch a drifting FT-100 or an open drain.	Instrument		OPEN	Risk 9 Medium
N6 Preheater E-100	R27	N6.4, N6.5	Confirm the E-100 shell design pressure $\geq$ the steam supply (11 bar a) or add PSV-110 on the shell; check the tube-rupture case (API 521 two-thirds rule).	Process		OPEN	Risk 8 Medium
N6 Preheater E-100	R28	N6.5, N6.6	Oil-in-condensate monitor on the E-100 condensate; condensate pot or trap by-pass check.	Instrument / Mechanical		OPEN	Risk 6 Medium
N6 Preheater E-100	R29	N6.1, N6.3, N6.7	Second outlet temperature element TE-110B with a deviation alarm, and TIC-110 in AUTO with set-point limits.	Instrument		OPEN	Risk 6 Medium
N7 D-150 and the cat-cracker interface	R30	N5.7, N7.1, N7.5	Give LSHH-150 its own transmitter and its own shut-off valve XV-150 at the D-150 inlet (today it reads LI-150 and acts through the control valve V100).	Instrument		OPEN	Risk 12 High

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N7 D-150 and the cat-cracker interface	R31	N1.6, N7.3	Confirm with the cat-cracker unit that the D-150 relief covers hot feed carrying water or light ends.	Process		OPEN	Risk 8 Medium
N8 Procedures	R32	N2.7, N4.3, N8.8, N8.9	Double block and bleed (or blinds) with a closed drain and vent for maintenance isolation of each pump; written isolation certificate.	Operations / Mechanical		OPEN	Risk 12 High
N8 Procedures	R33	N8.4, N8.7	Simulator training on the trip of the running pump, the change-over, a blocked discharge and back-flow through the idle pump; time the operator response against the D-150 hold-up.	Operations / Training		OPEN	Risk 12 High

HAZOP REPORT

Hazard and Operability Study of the cat-cracker feed pump station P-200A/B
Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator

	Role	Name	Signature	Date
Prepared by	HAZOP scribe			
Reviewed by	HAZOP leader (chair)			
Approved by	Plant / project manager			

This report records a teaching HAZOP of the pump station drawn in the simulator. Names, signatures and dates are filled in by the class team that runs the study; the risk matrix is an example for teaching, not a corporate matrix.

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Executive summary

A HAZOP of the cat-cracker feed pump station — the tank-farm transfer and storage tank T200, the common suction with strainer S-200, the running pump P-200A and the installed spare P-200B with their check and block valves, the discharge header with FE-100 / V100 under FRC-100, the steam preheater E-100 and the feed drum D-150 with the cat-cracker interface, and the start-up, change-over, shutdown and maintenance procedures — was carried out with the IEC 61882 guide-word method. The unit was divided into **8 nodes**; **64 deviations** were recorded and **33 recommendations** raised; **10 deviations** were also run on the simulator model (Section 5.3).

Nodes	Deviations	Recommendations	Very high	High	Medium	Low
8	64	33	0	10	51	3

Risk is ranked with the existing safeguards credited (Section 3.4). No deviation was ranked very high; the 10 high-ranked deviations (N3.1, N3.2, N3.8, N4.1, N5.1, N6.2, N7.1, N8.2, N8.4, N8.8) are those where a single trip, a mechanical check valve or the operator is the main protection — loss or deadhead of the running pump, back-flow through the idle spare, oil heated with no flow, D-150 overflow, and the procedures around the spare. They are carried to the LOPA, the SRS of the SIFs (PPS-V1-IN-SRS-001) and the cause and effect diagram (PPS-V1-IN-CE-001).

Key findings (dynamic check on the simulator)

- **Running pump trips (N3):** the motor-trip and FRC-100 LO alarms come at once; LI-150 LO follows at 07:10 and LALL-150 takes the cat cracker off at 14:19. That interval is the whole window for a manual start of the spare (R09). With the steam left on, TI-110 alarms at 00:18 and the oil in E-100 reaches 184 °C (R10).
- **Discharge valve closed on the running pump (N3):** TI-212 HI at 04:51, TSHH-212 stops the motor at 10:05 with the casing at 110 °C. The trip works, but only after the casing has been hot for minutes, and the feed is lost meanwhile — a low-flow trip and a minimum-flow line act earlier (R11, R12).
- **Check valve on the idle spare passing (N4):** the only annunciation in 30 min is FRC-100 LO; the feed settles at 10.6–11.2 L/s with V100 at 100 % and D-150 down to 33 %. Nothing trips and nothing points to the cause, so the failure stays hidden until someone looks for it (R18, R19, R20).
- **Strainer fouled (N2):** FRC-100 LO at once and the NPSH margin goes to –1.1 m — the pump cavitates — but no instrument in the control room shows the suction side (R07, R08).
- **Light ends (N2):** no alarm at all in 30 min, although the NPSH margin is –1.0 to 0.0 m and the flow sags to 12.9 L/s, just above the FRC-100 LO limit. Silent cavitation is a damage mechanism, not an upset the alarms see (R03, R08).
- **Transfer lost with T200 already low (N1):** LSLL-200 trips both pumps at 27:03. From the design level the same loss would leave about 4.5 h before the trip — T200 is a large buffer. The trip stops the motors only: V100 and TV-110 stay as they were, so TI-110 alarms 18 s later (R10).
- **Cat-cracker charge lost (N7):** LI-150 HI at 10:43, LSHH-150 closes V100 at 16:05 — but the pump keeps running against the closed valve: TI-212 HI at 20:34 and TSHH-212 at 25:48. One protective action creates the next hazard (deadhead), so LSHH-150 should also stop the running pump or a low-flow trip should (R12, R30).
- **Instrument air lost (N5):** V100 fails closed as designed — safe for D-150 — and deadheads the running pump: TI-212 HI at 04:30, TSHH-212 at 09:43, LALL-150 at 14:19. A fail-safe position is only safe for the hazard it was chosen for (R11, R12).
- **TV-110 wide open (N6):** TI-110 HI at 00:50, the outlet reaches 167 °C and the shell 9.4 bar a; the alarm is the only protection, which matches a Medium ranking for an operability deviation (R29).
- **Pumps stop with a passing check valve (N5):** D-150 drains back through E-100, V100 and the idle pump at up to 4.8 L/s: LI-150 LO at 05:20, LALL-150 at 10:39, D-150 down to 0 % and 8.7 m³ of hot oil back toward T200 in 30 min. Nothing in the design stops it (R22).

Actions by action party

Action party	Number	Action No.
Instrument	16	R01, R02, R07, R08, R09, R10, R12, R18, R19, R22, R24, R25, R26, R28, R29, R30
Mechanical	10	R05, R08, R11, R13, R15, R17, R20, R21, R28, R32
Operations	8	R02, R03, R04, R09, R16, R23, R32, R33
Process	6	R03, R06, R11, R21, R27, R31
Electrical	4	R14, R15, R16, R19

Action party	Number	Action No.
Safety	1	R06
Training	1	R33

An action shared by two parties (e.g. Instrument / Operations) is counted under both.

1 Introduction

1.1 Purpose

The HAZOP identifies how the pump station can deviate from its design intent, what the deviations lead to, whether the existing safeguards are adequate, and what should be changed. It also finds operability problems: deviations that harm no one but leave the cat cracker short of feed, damage a pump or upset the preheater. The study feeds the safety layers, LOPA and SRS of the SIFs, and the cause and effect diagram PPS-V1-IN-CE-001.

1.2 The unit

Gas oil at 60 °C is drawn from the cone-roof tank T200 (design level 30 %, 339 m³ ≈ 6.7 h of feed) through the common suction valve V200 and strainer S-200 by two identical centrifugal pumps in parallel, P-200A (running) and P-200B (installed spare, lined up), each with suction valve, local casing gauge, swing check valve and discharge block valve. At design pump A delivers 14.0 L/s with a casing head of 32.4 m, 13.2 A and an NPSH margin of 7.8 m. FRC-100 (remote set point from the cat cracker) holds the feed with V100 at 84 % (fail closed). The steam preheater E-100 (TV-110 in manual, 4.0 t/h of steam, shell 6.0 bar a) heats the feed to 150 °C into the feed drum D-150 (30 m³, 50 %), from which the cat cracker draws its charge. Protection: LSL-200 (10 %, 5 s) trips both pumps; TSHH-212/222 (110 °C) and the overload relays 49-210/220 (21.7 A, 10 s) stop one motor; LSHH-150 (95 %) forces V100 shut; the cat cracker trips its own charge pump on LALL-150 (10 %); an ESD push-button stops both pumps and closes V100 and TV-110.

1.3 Objectives

- Identify hazards to people, the environment and the plant from deviations of each node and each procedure step.
- Identify operability problems: feed short of the cat-cracker demand, damaged pumps and seals, an unready spare, off-temperature feed.
- Judge the existing safeguards and rank the risk with the example matrix.
- Raise recommendations and pass the high-ranked scenarios to LOPA.
- Check selected deviations dynamically on the simulator model (Section 5.3).

2 Scope

Boundaries: scope from the tank-farm transfer valve (HIC-199) through T200, the suction, P-200A/B with their check and block valves, the discharge header, FE-100/V100, E-100 (steam and feed sides) to the D-150 inlet and the interface with the cat cracker (remote set point, LALL-150); excluded: the tank farm, the steam system upstream of TV-110 and the cat-cracker unit itself. Objectives, team, risk matrix.

2.1 Nodes

Node	Description	Design intent	Design conditions	Equipment / instruments
N1	Tank-farm transfer and T200	Receive gas oil at 60 °C from the tank farm through HIC-199 (100 % = 20 L/s) into cone-roof tank T200 (Ø 12 m × 10 m, 113.1 m ³) and keep the level near 30 % (3 m) as the suction head and buffer for the pumps.	Level 30 % = 339 m ³ ≈ 6.7 h of feed (4.5 h to LSL-200); transfer 14.0 L/s in = out; atmospheric, 60 °C	Tank-farm transfer line and HIC-199 valve, T200 (cone roof, vent), outlet nozzle to V200 <i>HIC-199, FI-199, LI-200, LSL-200 (permissive), LSL-200 (trip), TI-200</i>
N2	Suction: V200, S-200 and the suction header	Deliver liquid from T200 through V200 and strainer S-200 to the pump suctions with a small loss (3.5 m at 14 L/s), leaving NPSH available well above the NPSH required.	Suction head ≈ 0.5 m at the header; NPSH available 12.3 m vs required 4.5 m on the running pump (margin 7.8 m); 14.0 L/s	T200 outlet line, V200 (gate), S-200 strainer, common suction header, V202 / V206 <i>PI-209 (local gauge only)</i>
N3	Pump A leg (P-200A, normally running)	P-200A (3600 rpm, 10 kW motor, shut-off 45 m) raises the gas oil from the suction header to about 32 m casing head at 14 L/s and delivers it through NRV-210 and V201 into the discharge header.	14.0 L/s, casing 32.4 m, 5.72 kW, 13.2 A (FLA 18.9 A), casing 60 °C, NPSH margin 7.8 m	V202, P-200A and motor, vent VA, NRV-210, V201 <i>PI-210 (local), II-210, TI-212, TSHH-212, 49-210 overload relay</i>

Node	Description	Design intent	Design conditions	Equipment / instruments
N4	Pump B leg (P-200B, spare) with the check valves	P-200B (shut-off 46.5 m) stands by lined up — V206 and V205 open, casing vented and full — ready for a manual start; NRV-220 keeps the header pressure off the idle pump.	Idle: casing at the suction head (0.5 m), 53 °C, no flow; header 32.0 m held back by NRV-220	V206, P-200B and motor, vent VB, NRV-220, V205 <i>PI-220 (local), II-220, TI-222, TSHH-222, 49-220 overload relay</i>
N5	Discharge header, FE-100, V100 and FRC-100	Carry the pump discharge (about 32 m) through FE-100 and V100 (modified equal percentage, fail closed) at the 14 L/s asked for by the cat cracker; FRC-100 (PI, remote set point) holds the flow with V100 near 84 %.	14.0 L/s at 32.0 m header head; V100 84 %; FRC-100 Kc 0.35, Ti 3 s, span 0–20 L/s	Common discharge header, low-point drain DRN, FE-100, V100 <i>FRC-100 (FT-100), V100 positioner</i>
N6	Preheater E-100: steam side and feed side	Heat the feed from 60 to 150 °C (about 2.3 MW) with steam supplied at 11 bar a through TV-110 (fail closed; TIC-110 normally in MAN near 50 %); shell near 6 bar a, condensate through a trap.	2.31 MW, steam 4.03 t/h, shell 5.97 bar a (159 °C), outlet 150 °C, TV-110 50 %	E-100 (steam in the shell), TV-110, steam trap, condensate return <i>TIC-110 (TI-110), PI-111, FI-112</i>
N7	D-150 and the cat-cracker interface	Hold up to 30 m ³ of hot feed (design 50 %, 1.5 bar g) as surge between the station and the cat cracker, which draws 14 L/s (FI-151); the station does not control the D-150 level — the cat cracker resets FRC-100 remotely.	50 % = 15 m ³ ; 14.3 min to LALL-150 with no feed, 16.1 min to LSHH-150 with no charge (at 14 L/s)	D-150, charge line to the cat cracker <i>LI-150, LSHH-150, LALL-150 (cat-cracker side), FI-151</i>
N8	Procedures: start-up, change-over, shutdown, maintenance of the spare	Start a pump vented, against a closed discharge, open it within 2 min; establish flow before steam; change over A → B by starting B against a closed V205; shut down steam before flow; isolate, drain and vent the spare safely for maintenance.	Shut-off check ≥ 40 m on the local gauge before opening the discharge; deadhead heating about 6 °C/min; 14.3 min of D-150 hold-up from 50 % to LALL-150	P-200A/B, V201/V205, V202/V206, vents VA/VB, V100, TV-110 <i>PI-210/220 (local), FRC-100, TIC-110, II-210/220, TI-212/222</i>

2.2 Node map

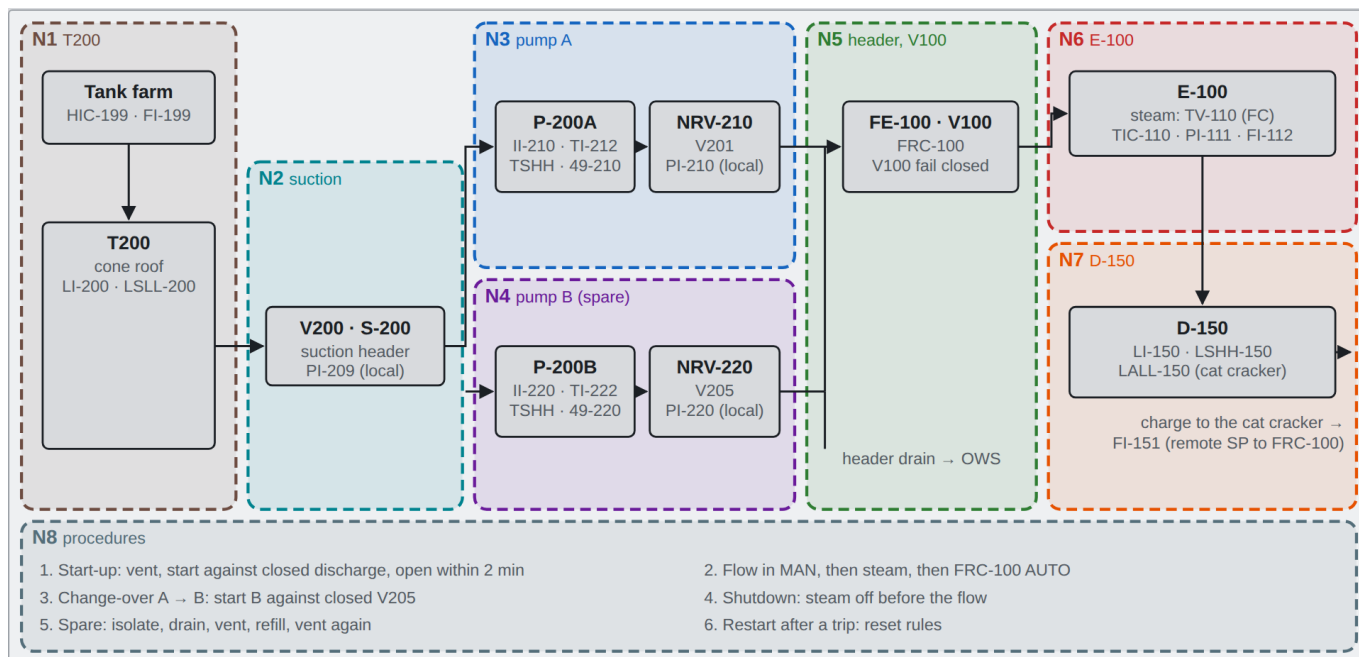


Figure 1. The pump station divided into HAZOP nodes (N8, the procedures node, covers the whole station).

3 Methodology

HAZOP is a structured, team-based examination of a defined system. Each part of the design is compared with its **design intent**; **guide words** combined with process **parameters** generate **deviations**; for each credible deviation the team records causes, consequences, existing safeguards and, where the risk is not acceptable, recommendations. It finds both hazards (harm to people, environment, assets) and operability problems (the station cannot deliver the feed the cat cracker asks for). On a pump station with an installed spare, the hazards sit as much in the idle equipment — a check valve that no longer seats, a casing blocked in for maintenance — as in the running pump, so **REVERSE** (back-flow through an idle pump, a pump turning backwards) and the procedure node matter as much as NO flow.

3.1 References

Reference	What it contributes
IEC 61882:2016 <i>Hazard and operability studies (HAZOP studies) – Application guide</i>	Definitions, the four phases (definition, preparation, examination, documentation and follow-up), guide words, team roles, recording
CCPS <i>Guidelines for Hazard Evaluation Procedures</i> (3rd ed.)	Worksheet format, node selection, procedural HAZOP, "consequence without safeguards", risk ranking, links to LOPA
IEC 61511 (functional safety, SIS)	How HAZOP scenarios feed LOPA and SIL targets for the trips LSHH-200, TSHH-212/222 and LSHH-150 and for proposed ones (low-flow trip, overfill protection)
API 610 (centrifugal pumps), API 682 (seals), API 2350 (tank overfill), API 2000 (tank venting)	Minimum continuous flow, reverse rotation, seal arrangements, overfill and vacuum protection used as the basis for safeguards
ISO 31000 / owner's risk matrix	Risk criteria used to decide whether safeguards are adequate
Thailand: Department of Industrial Works regulation on hazard identification, risk assessment and risk-management plans (B.E. 2543)	Legal requirement to identify hazards (HAZOP is an accepted method) and to prepare risk-management plans for factories
OSHA 29 CFR 1910.119 (PSM), where applicable	Process hazard analysis for covered processes, revalidated at least every five years

3.2 The four phases

- **Definition:** scope from the tank-farm transfer valve (HIC-199) through T200, the suction, P-200A/B with their check and block valves, the discharge header, FE-100/V100, E-100 (steam and feed sides) to the D-150 inlet and the interface with the cat cracker (remote set point, LALL-150); excluded: the tank farm, the steam system upstream of TV-110 and the cat-cracker unit itself. Objectives, team, risk matrix.
- **Preparation:** collect the P&ID, the pump curves, the process and design data, the trip and alarm list, the start-up, change-over and shutdown procedures, datasheets; divide the plant into nodes; write each node's design intent; choose parameters and guide words.
- **Examination:** for each node → state the design intent → choose a parameter → apply each guide word → is the deviation meaningful? → causes → consequences assuming the safeguards fail → existing safeguards → rank the risk → recommend → next guide word. Record every meaningful deviation, including those with adequate safeguards ("full recording"), so the reasoning can be audited.
- **Documentation and follow-up:** worksheets, recommendation register with owners and dates, action close-out, and revalidation after changes (MOC) or periodically.

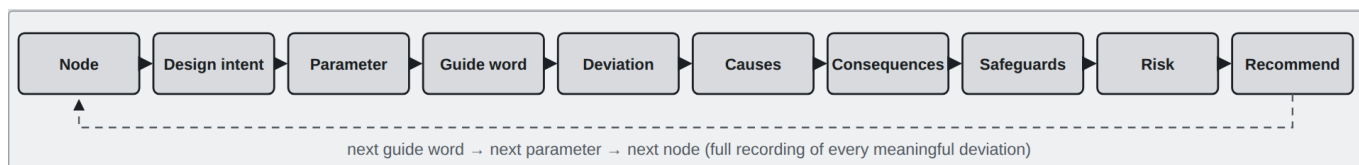


Figure 2. HAZOP examination loop.

3.3 Guide words and parameters

Guide word	Meaning (IEC 61882)	Rows
NO / NOT	Complete negation of the intent	8
MORE	Quantitative increase	18
LESS	Quantitative decrease	9
AS WELL AS	Qualitative increase: something extra present	5
PART OF	Qualitative decrease: only part of the intent achieved	4

Guide word	Meaning (IEC 61882)	Rows
REVERSE	Logical opposite of the intent	4
OTHER THAN	Complete substitution: something else happens	9
EARLY	Relative to clock time	1
LATE	Relative to clock time	2
BEFORE	Relative to order or sequence	3
AFTER	Relative to order or sequence	1

Parameters used: flow, pressure, temperature, level, composition, rotation, current, integrity (containment), measurement, control, and for procedures: sequence, timing, maintenance, electrical work, restart. A deviation is recorded only if it is physically meaningful ("REVERSE temperature" is not; REVERSE flow and REVERSE rotation are).

3.4 Risk ranking

S \ L	L1	L2	L3	L4	L5
S5	5	10	15	20	25
S4	4	8 (14)	12 (6)	16	20
S3	3	6 (10)	9 (13)	12 (4)	15
S2	2	4 (3)	6 (10)	8 (4)	10
S1	1	2	3	4	5

Numbers in brackets: how many deviations of this study fall in the cell. Risk = S × L (the same matrix ranks the HAZID register).

S	Severity (worst credible consequence, no credit for safeguards)
1	No injury; short feed upset
2	First aid; feed off target or a pump damaged; small contained spill
3	Lost-time injury; cat cracker loses its feed (unit trip); pump out for days; local release
4	Permanent disability or single fatality; gas-oil fire at the pumps or the tank; major equipment damage
5	Multiple fatalities; tank fire spreading beyond the bund; escalation to the cat cracker

L	Likelihood with the existing safeguards
1	Remote: not expected in the life of the plant ($< 10^{-4}$ /y)
2	Unlikely: known in industry (10^{-4} – 10^{-3} /y)
3	Possible: could happen in the plant life (10^{-3} – 10^{-2} /y)
4	Likely: several times in the plant life (10^{-2} – 10^{-1} /y)
5	Frequent: yearly or more

1–4 Low acceptable · 5–9 Medium reduce if practicable (ALARP), review safeguards · 10–16 High recommendation required, consider LOPA · 20–25 Very high not acceptable, act before operation

4 HAZOP sessions

4.1 Schedule

The study is run in class as a teaching HAZOP. A typical plan is three sessions; the team records the dates and attendance on this page.

Session	Nodes	Date	Remark
1	N1 – N3: Tank-farm transfer and T200; Suction; Pump A leg		
2	N4 – N6: Pump B leg; Discharge header, FE-100, V100 and FRC-100; Preheater E-100		
3	N7 – N8: D-150 and the cat-cracker interface; Procedures; review of the recommendations		

4.2 Team

Role	Responsibility	Name	Attended sessions
HAZOP leader (chair)	Independent of the design; runs the method, keeps the team on the node, decides when a deviation is covered		
Scribe (recorder)	Writes the worksheet live, reads back recommendations		
Process engineer (designer)	Design intent, hydraulics (system and pump curves, NPSH), heat balance of E-100		
Operations representative	How the station is really run: rounds, the spare on standby, change-over, the cat-cracker interface		
Instrument / control engineer	Loops, fail positions, trips and permissives, which transmitter each trip reads		
Safety / risk engineer	Risk matrix, LOPA follow-up, regulations		
Specialists as needed	Rotating equipment (pumps, seals, check valves), electrical (motor protection, phase sequence), cat-cracker operations		

4.3 Documents reviewed

Document	Title / content
PPS-PID-001 rev A	P&ID of the cat-cracker feed pumps P-200A/B, T200, E-100, D-150 (EPC sheet, P&ID tab; Appendix 2)
PPS-V1-PID-001	The same unit in the DCS layout used by the console and the exercises (P&ID tab; Appendix 2)
PPS-PFD-100	Process flow diagram with stream numbers 1-11 (PFD tab; Appendix 2)
PPS-V1-IN-CE-001 rev R0	Cause and effect diagram of the SIS, motor protection, permissives and alarms (C&E matrix tab)
PPS-V1-IN-SRS-001 rev R0	Safety requirements specification of the SIFs (SRS of SIF tab)
Simulator tabs	Process (stream table, balances), Pump curves (curves, NPSH), Design of the unit operation, Control loop design, SIS (trips, permissives, Rev C options), Workbook (alarm list), start-up / change-over / shutdown procedures (Exercises)

4.4 Nodes reviewed and guide-word coverage

Node	NO / NOT	MORE	LESS	AS WELL AS	PART OF	REVERSE	OTHER THAN	EARLY	LATE	BEFORE	AFTER	Rows	Parameters
N1	1	2	2	2	·	·	1	·	·	·	·	8	Flow, Level, Composition, Pressure, Integrity
N2	2	1	2	1	1	·	·	·	·	·	·	7	Flow, Pressure, Composition, Measurement
N3	3	3	1	·	·	1	1	·	·	·	·	9	Flow, Rotation, Current, Temperature, Pressure, Integrity, Measurement
N4	1	2	1	·	·	2	·	·	·	·	·	6	Flow, Rotation, Pressure, Temperature
N5	1	3	1	1	1	1	1	·	·	·	·	9	Flow, Pressure, Control
N6	·	4	1	·	·	·	2	·	·	·	·	7	Temperature, Pressure, Integrity, Level, Measurement
N7	·	3	1	·	·	·	2	·	·	·	·	6	Level, Pressure, Flow, Measurement, Control
N8	·	·	·	1	2	·	2	1	2	3	1	12	Sequence, Timing, Maintenance, Electrical work, Restart

A dot means that guide word gave no meaningful deviation for any parameter in the node (it was still considered). EARLY / LATE / BEFORE / AFTER apply mainly to the procedure node N8.

5 Findings and recommendations

5.1 Risk profile

Node	Deviations	High	Medium	Low	Recommendations
N1 Tank-farm transfer and T200	8	0	8	0	R01, R02, R03, R04, R05, R06
N2 Suction	7	0	7	0	R03, R07, R08
N3 Pump A leg	9	3	5	1	R06, R08, R09, R10, R11, R12, R13, R14, R16, R17, R18
N4 Pump B leg	6	1	4	1	R13, R15, R16, R17, R18, R19, R20, R21
N5 Discharge header, FE-100, V100 and FRC-100	9	1	8	0	R11, R12, R21, R22, R23, R24, R25, R26
N6 Preheater E-100	7	1	6	0	R10, R27, R28, R29
N7 D-150 and the cat-cracker interface	6	1	5	0	R04, R09, R12, R26, R30, R31
N8 Procedures	12	3	8	1	R09, R10, R11, R14, R15, R16, R19, R21, R32, R33
Total	64	10	51	3	33

High-ranked deviations

Ref	Deviation	Consequence (no credit for safeguards)	S×L	Rec.
N3.1	Flow NO: Running pump stops	Feed to the cat cracker lost; D-150 empties and the cat cracker trips its charge pump; with the steam still on, the oil standing in E-100 heats toward the steam temperature.	3×4 = 12	R09, R10
N3.2	Flow NO: Pump running against a blocked discharge (deadhead)	All motor power goes into the casing liquid (about 6 °C per minute): seal faces flash, the seal fails, hot gas oil is released; prolonged deadhead can damage the casing.	4×3 = 12	R11, R12, R24
N3.8	Integrity OTHER THAN: Seal leak	Gas oil at 60 °C sprays or pools under the pump; ignition on the hot motor or coupling gives a pool fire that can spread to the spare.	4×3 = 12	R06, R17
N4.1	Flow REVERSE: Back-flow through the idle pump	Part of the running pump flow circulates back through the idle pump to the suction: the feed to the cat cracker is short with V100 wide open, D-150 falls, the idle pump is driven backwards.	3×4 = 12	R18, R19, R20
N5.1	Flow NO: No flow through V100	Running pump deadheaded (see N3) and the cat cracker loses its feed; D-150 empties.	4×3 = 12	R11, R12
N6.2	Temperature MORE: Oil heated with no flow	The oil standing in the tubes heats toward the steam temperature (184 °C at the supply pressure): coking and fouling of the tubes; a slug of overheated oil at the restart.	3×4 = 12	R10
N7.1	Level MORE: D-150 fills	D-150 overfills: hot oil to the cat-cracker vent or relief, liquid carry-over; if V100 is then closed, the running pump is deadheaded.	4×3 = 12	R12, R30
N8.2	Sequence BEFORE: Pump started before it is vented or with the suction closed	Dry running: the seal fails in about a minute and sprays gas oil; fire.	4×3 = 12	R16
N8.4	Timing LATE: Spare not started in time after a trip	D-150 runs down and LALL-150 trips the cat cracker.	3×4 = 12	R09, R33
N8.8	Maintenance OTHER THAN: Spare opened while still under pressure	Gas oil at about 60 °C and header pressure released onto the fitter: injury, fire.	4×3 = 12	R32, R21

5.2 Recommendations

No.	Recommendation	Node	Raised from	Action by
R01	Give LSL-200 its own level transmitter LT-200B (today the LI-200 alarm, the LSL-200 start permissive and the LSL-200 trip read one measurement).	N1	N1.1, N1.3	Instrument
R02	Independent high-high level LSHH-200 on T200 that stops the tank-farm transfer (overflow protection to API 2350).	N1	N1.2, N1.4	Instrument / Operations
R03	Feed quality at the tank farm: vapour-pressure / flash-point check of each transfer and a line-up check, so naphtha or light material cannot reach T200 unseen.	N1/N2	N1.5, N2.3	Operations / Process

No.	Recommendation	Node	Raised from	Action by
R04	T200 water draw-off routine (each shift) and water detection at the tank outlet before the feed is heated in E-100.	N1/N7	N1.6, N7.3	Operations
R05	Confirm the T200 pressure/vacuum vent (API 2000) for the largest pump-out and fill rates; inspect the flame arrester.	N1	N1.7	Mechanical
R06	Remotely operated, fire-safe emergency isolation valve EIV-200 on the T200 outlet, closable from the control room.	N1/N3	N1.8, N3.8	Process / Safety
R07	Suction pressure transmitter PIT-209 with a low alarm in the DCS, and a differential-pressure gauge across S-200.	N2	N2.1, N2.2, N2.6	Instrument
R08	Vibration transmitters on P-200A/B with DCS alarms (VI-210/220 are local today).	N2/N3	N1.5, N2.2, N2.3, N3.9	Instrument / Mechanical
R09	Auto-start of the spare pump on loss of the running pump (motor trip or low header pressure), with the zero-speed permissive of R15.	N3/N7/N8	N3.1, N4.4, N5.2, N7.2, N8.4	Instrument / Operations
R10	Steam cut-off: close TV-110 automatically when the feed flow is lost (FT-100 low or both pumps stopped).	N3/N6/N8	N3.1, N6.2, N8.1, N8.6	Instrument
R11	Minimum-flow recycle from each pump discharge (upstream of the check valve) back to T200, sized for the API 610 minimum continuous flow.	N3/N5/N8	N3.2, N4.5, N5.1, N8.3	Process / Mechanical
R12	Low-flow trip of the running pump (flow in each pump leg, or FT-100 low with V100 closed, time-delayed) so a blocked discharge stops the motor before the casing gets hot.	N3/N5/N7	N3.2, N4.5, N5.1, N7.1	Instrument
R13	Pump condition monitoring: performance test against the curve every quarter, wear-ring clearances at overhaul.	N3/N4	N2.5, N3.3	Mechanical
R14	Rotation (bump) test and phase-sequence check in the close-out of any electrical work on the pump motors.	N3/N8	N3.4, N8.11	Electrical
R15	Zero-speed / reverse-rotation switch as a start permissive for each pump (or a non-reverse ratchet on the coupling).	N4/N8	N4.2, N8.7	Electrical / Mechanical
R16	Dry-run protection: under-power relay on each motor, and a vent sight glass to prove the casing is liquid-full before a start.	N3/N4/N8	N2.1, N2.4, N3.6, N4.4, N8.2, N8.10	Electrical / Operations
R17	Dual mechanical seals (API 682) with seal-leak detection; seal drip pans with a kerb to the oily-water sewer.	N3/N4	N3.8, N8.9	Mechanical
R18	Casing pressure transmitters PIT-210/220 with a high alarm on the idle pump (Rev C option on the SIS tab).	N3/N4	N3.9, N4.1	Instrument
R19	MOV discharge valves V201/V205 interlocked with the motors (Rev C option): open after the start, close on every stop.	N4/N8	N4.1, N4.2, N8.7	Instrument / Electrical
R20	Check valves: non-slam type in an inspection and seat-test programme; consider a second check valve in series on each leg.	N4	N4.1, N5.5	Mechanical
R21	Thermal relief valve on each blocked-in liquid section: pump casing / discharge between check valve and block valve, and the header between the check valves and V100.	N4/N5/N8	N2.7, N4.3, N4.6, N5.8, N8.6, N8.8	Process / Mechanical
R22	Close V100 automatically (FRC-100 to MAN 0 %) when both pumps are stopped, so D-150 cannot drain back through a passing check valve.	N5	N5.5	Instrument
R23	Plug or lock the header low-point drain; keep a lock / car-seal register of drains and vents.	N5	N5.6	Operations
R24	Header pressure transmitter PIT-100 upstream of FE-100/V100 with a high alarm (blocked outlet, pump at shut-off) and a low alarm (pump lost).	N5	N3.2, N3.7	Instrument
R25	FRC-100 output-high alarm (V100 above 95 % = loop saturated) and a high clamp on the remote set point.	N5	N3.3, N5.2, N5.4	Instrument
R26	Flow-balance alarm FT-100 against FI-151 and the D-150 level trend, to catch a drifting FT-100 or an open drain.	N5/N7	N5.3, N5.6, N7.2, N7.4, N7.6	Instrument
R27	Confirm the E-100 shell design pressure $\geq$ the steam supply (11 bar a) or add PSV-110 on the shell; check the tube-rupture case (API 521 two-thirds rule).	N6	N6.4, N6.5	Process
R28	Oil-in-condensate monitor on the E-100 condensate; condensate pot or trap by-pass check.	N6	N6.5, N6.6	Instrument / Mechanical
R29	Second outlet temperature element TE-110B with a deviation alarm, and TIC-110 in AUTO with set-point limits.	N6	N6.1, N6.3, N6.7	Instrument
R30	Give LSHH-150 its own transmitter and its own shut-off valve XV-150 at the D-150 inlet (today it reads LI-150 and acts through the control valve V100).	N7	N5.7, N7.1, N7.5	Instrument
R31	Confirm with the cat-cracker unit that the D-150 relief covers hot feed carrying water or light ends.	N7	N1.6, N7.3	Process

No.	Recommendation	Node	Raised from	Action by
R32	Double block and bleed (or blinds) with a closed drain and vent for maintenance isolation of each pump; written isolation certificate.	N8	N2.7, N4.3, N8.8, N8.9	Operations / Mechanical
R33	Simulator training on the trip of the running pump, the change-over, a blocked discharge and back-flow through the idle pump; time the operator response against the D-150 hold-up.	N8	N8.4, N8.7	Operations / Training

5.3 Dynamic check of selected deviations on the simulator

A conventional HAZOP judges consequences qualitatively. Here ten deviations are run on copies of the pump-station model from the design state (P-200A running at 14 L/s, B lined up, FRC-100 in AUTO, TIC-110 in MAN) with the existing safeguards active — alarms, SIS and motor-protection trips, controllers in their normal modes, tank and drum levels free to move — and **no operator action**, for 30 min. The result shows whether the safeguards act before harm and how much time the operator has. Alarms are checked every second with the page's alarm table and its suppression rules; the live simulator is not touched.

Ref	Node	Deviation	Initiating event (on the model)	First alarms (mm:ss)	Trips and protective actions	Key tags: range in 30 min
D1	N3	Flow NO	Running pump A motor trips; the spare is not started	00:00 FRC-100 LO 00:00 P-200A TRIP 00:18 TI-110 HI 07:10 LI-150 LO 14:19 FI-151 LOST	P-200A stopped by the initiating event at 00:00 LALL-150 trips the cat-cracker charge at 14:19	- Feed to E-100 (real) 0.0 to 12.4 L/s - LI-150 10 to 50 % - TI-110 150 to 184 °C
D2	N3	Flow NO (deadhead)	Discharge block valve V201 closed in the field while A runs	00:23 FRC-100 LO 00:41 TI-110 HI 04:51 TI-212 HI 07:33 LI-150 LO 10:05 P-200A TRIP	TSHH-212 trips P-200A at 10:05 LALL-150 trips the cat-cracker charge at 14:42	- TI-212 58 to 110 °C - Feed to E-100 (real) 0.0 to 14.0 L/s - LI-150 10 to 50 %
D3	N4	Flow REVERSE	Check valve on the idle spare stops seating	00:00 FRC-100 LO	no trip	- Feed to E-100 (real) 10.6 to 11.2 L/s - V100 84 to 100 % open - LI-150 33 to 50 %
D4	N2	Pressure LESS	Suction strainer S-200 heavily fouled (line loss x 8)	00:00 FRC-100 LO 01:27 TI-110 HI 22:37 LI-150 LO	no trip	- Feed to E-100 (real) 9.5 to 10.0 L/s - NPSH margin, pump A –1.1 to 0.5 m - LI-150 23 to 50 %
D5	N2	Composition AS WELL AS	Light ends in the feed (vapour pressure toward 75 kPa)	none in 30 min	no trip	- NPSH margin, pump A –1.0 to 0.0 m - Feed to E-100 (real) 12.9 to 14.0 L/s - FRC-100 (FT-100) 12.9 to 14.0 L/s
D6	N1	Flow NO (transfer)	Tank-farm transfer lost with T200 already down at 12 %	00:00 LI-200 LO 27:03 FRC-100 LO 27:03 SIS TRIP 27:21 TI-110 HI	LSLL-200 T200 level low-low unit trip at 27:03	- LI-200 10.0 to 12.0 % - Feed to E-100 (real) 0.0 to 14.0 L/s - LI-150 42 to 50 %
D7	N7	Level MORE	Cat-cracker charge pump trips; the station keeps feeding	10:43 LI-150 HI 16:05 LSHH-150 TRIP 16:24 TI-110 HI 20:34 TI-212 HI 25:48 P-200A TRIP	TSHH-212 trips P-200A at 25:48 LSHH-150 closes V100 at 16:05	- LI-150 50 to 95 % - V100 0 to 84 % open - TI-212 60 to 110 °C
D8	N5	Flow NO (air)	Instrument-air failure: V100 and TV-110 go to their fail positions	00:01 FRC-100 LO 04:30 TI-212 HI 07:10 LI-150 LO 09:43 P-200A TRIP 14:19 FI-151 LOST	TSHH-212 trips P-200A at 09:43 LALL-150 trips the cat-cracker charge at 14:19	- TI-212 58 to 110 °C - V100 0 to 74 % open - LI-150 10 to 50 %
D9	N6	Temperature MORE	TIC-110 output driven to 100 % (TV-110 wide open)	00:50 TI-110 HI	no trip	- TI-110 150 to 167 °C - E-100 shell 8.2 to 9.4 bar a - Feed to E-100 (real) 14.0 to 14.0 L/s
D10	N5	Flow REVERSE	Power dip stops the running pump while the spare check valve passes	00:00 FRC-100 LO 00:18 TI-110 HI 05:20 LI-150 LO 10:39 FI-151 LOST	LALL-150 trips the cat-cracker charge at 10:39	- Feed to E-100 (real) –4.8 to 9.1 L/s - LI-150 0 to 50 % - TI-110 150 to 184 °C

What the runs tell the HAZOP team

• **Running pump trips (N3):** the motor-trip and FRC-100 LO alarms come at once; LI-150 LO follows at 07:10 and LALL-150 takes the cat cracker off at 14:19. That interval is the whole window for a manual start of the spare (R09). With the steam left on, TI-110 alarms at 00:18 and the oil in E-100 reaches 184 °C (R10).

• **Discharge valve closed on the running pump (N3):** TI-212 HI at 04:51, TSHH-212 stops the motor at 10:05 with the casing at 110 °C. The trip works, but only after the casing has been hot for minutes, and the feed is lost meanwhile — a low-flow trip and a minimum-flow line act earlier (R11, R12).

- **Check valve on the idle spare passing (N4):** the only annunciation in 30 min is FRC-100 LO; the feed settles at 10.6–11.2 L/s with V100 at 100 % and D-150 down to 33 %. Nothing trips and nothing points to the cause, so the failure stays hidden until someone looks for it (R18, R19, R20).
- **Strainer fouled (N2):** FRC-100 LO at once and the NPSH margin goes to –1.1 m — the pump cavitates — but no instrument in the control room shows the suction side (R07, R08).
- **Light ends (N2):** no alarm at all in 30 min, although the NPSH margin is –1.0 to 0.0 m and the flow sags to 12.9 L/s, just above the FRC-100 LO limit. Silent cavitation is a damage mechanism, not an upset the alarms see (R03, R08).
- **Transfer lost with T200 already low (N1):** LSLL-200 trips both pumps at 27:03. From the design level the same loss would leave about 4.5 h before the trip — T200 is a large buffer. The trip stops the motors only: V100 and TV-110 stay as they were, so TI-110 alarms 18 s later (R10).
- **Cat-cracker charge lost (N7):** LI-150 HI at 10:43, LSHH-150 closes V100 at 16:05 — but the pump keeps running against the closed valve: TI-212 HI at 20:34 and TSHH-212 at 25:48. One protective action creates the next hazard (deadhead), so LSHH-150 should also stop the running pump or a low-flow trip should (R12, R30).
- **Instrument air lost (N5):** V100 fails closed as designed — safe for D-150 — and deadheads the running pump: TI-212 HI at 04:30, TSHH-212 at 09:43, LALL-150 at 14:19. A fail-safe position is only safe for the hazard it was chosen for (R11, R12).
- **TV-110 wide open (N6):** TI-110 HI at 00:50, the outlet reaches 167 °C and the shell 9.4 bar a; the alarm is the only protection, which matches a Medium ranking for an operability deviation (R29).
- **Pumps stop with a passing check valve (N5):** D-150 drains back through E-100, V100 and the idle pump at up to 4.8 L/s: LI-150 LO at 05:20, LALL-150 at 10:39, D-150 down to 0 % and 8.7 m³ of hot oil back toward T200 in 30 min. Nothing in the design stops it (R22).

5.4 Follow-up and close-out

- Each recommendation gets an owner and a due date; the owner answers *accepted and done*, *accepted with an alternative* (with justification) or *rejected* (with a justification agreed by the HAZOP leader and the plant manager). The register in section 8 is the tracking list.
- High scenarios whose protection relies on instrumented trips — the blocked discharge (TSHH-212/222), D-150 overfill (LSHH-150 through V100), loss of suction (LSLL-200 on the shared LI-200) — go to **LOPA** to decide whether the trips are enough or a new SIF (low-flow trip, overfill protection) is needed (IEC 61511). See the Safety layers and LOPA tabs.
- The Rev C options (casing pressure transmitters with PAH on the idle pump, MOV discharge valves interlocked with the motors) answer R18 and R19; putting them in service is a change and goes through **management of change**, which checks whether this HAZOP must be updated.
- Revalidate the HAZOP periodically (typically every five years) and after incidents or near misses.
- Use the simulator to train operators on the scenarios this HAZOP found and to time their response against the D-150 hold-up (R33).

Appendix 1 · HAZOP worksheets

Node N1 · Tank-farm transfer and T200

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Node	N1 · Tank-farm transfer and T200
Design intent	Receive gas oil at 60 °C from the tank farm through HIC-199 (100 % = 20 L/s) into cone-roof tank T200 (Ø 12 m × 10 m, 113.1 m³) and keep the level near 30 % (3 m) as the suction head and buffer for the pumps.	Design conditions	Level 30 % = 339 m³ ≈ 6.7 h of feed (4.5 h to LSL-200); transfer 14.0 L/s in = out; atmospheric, 60 °C
Equipment	Tank-farm transfer line and HIC-199 valve, T200 (cone roof, vent), outlet nozzle to V200	Instruments	HIC-199, FI-199, LI-200, LSL-200 (permissive), LSL-200 (trip), TI-200
Drawing	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Document	PPS-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.1	Flow	NO	No transfer into T200	- Tank-farm transfer pump stops (fault library: transfer lost) - HIC-199 set to 0 or its valve fails closed - block valve at the tank farm closed	T200 runs down while the pumps draw; at low level the pumps lose suction (vortex, air), run dry, the seals fail and hot gas oil is released; the cat cracker loses its feed.	- LI-200 LO 20 % / HI 85 % - LSLL-200 at 10 % for 5 s → both pumps tripped (SIS, first-out) - FI-199 indication - Hold-up: hours of feed at design	3	2	6 M	R01	Instrument
N1.2	Flow	MORE	Transfer larger than the draw	- HIC-199 left at 100 % (20 L/s) when the feed rate falls - the cat cracker trips and the station is cut back	Level rises until T200 overflows through the roof vents: gas oil pool in the tank area, fire if ignited.	- LI-200 LO 20 % / HI 85 % - Operator rounds at the tank	4	2	8 M	R02	Instrument / Operations
N1.3	Level	LESS	Level low with the instruments blind	- LI-200 transmitter sticks or reads high - one measurement serves the LI-200 alarm, LSL-200 and LSL-200	No alarm and no trip while the real level falls: both pumps run dry, seals fail, gas oil is released at the pumps (fire).	- FI-199 vs FRC-100 mismatch seen by the operator - Field level gauge on rounds (assumed)	4	2	8 M	R01	Instrument
N1.4	Level	MORE	Overfill of T200	- LI-200 reads low - transfer left on during a feed outage	Overflow of gas oil at about 60 °C into the tank area; vapour near the flash point; pool fire.	- LI-200 LO 20 % / HI 85 % - Tank-farm operator watches the transfer	4	2	8 M	R02	Instrument / Operations
N1.5	Composition	AS WELL AS	Light ends (naphtha) in the feed	- Wrong tank lined up at the tank farm - light material left in the line (fault library: light ends in the feed)	Vapour pressure rises toward 75 kPa, NPSH available falls by about 8 m: the pumps cavitate (erosion, vibration, seal damage); flammable vapour in the T200 roof space.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - Local vibration and noise at the pumps on rounds	4	2	8 M	R03, R08	Instrument / Mechanical · Operations / Process
N1.6	Composition	AS WELL AS	Water in the feed	Water settled in the tank-farm tank or in T200 drawn with the gas oil	Water heated in E-100 flashes in D-150 (water boils at about 128 °C at 1.5 bar g): pressure surge in D-150 and steam carried to the cat cracker.	T200 water draw-off by procedure (assumed)	4	2	8 M	R04, R31	Operations · Process
N1.7	Pressure	LESS	Vacuum in T200	P/V vent or flame arrester blocked while the pumps draw 14–20 L/s	Cone-roof tank pulled under vacuum: roof or shell buckles, the tank leaks.	Tank vent (not shown on PPS-PID-001)	3	2	6 M	R05	Mechanical

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.8	Integrity	OTHER THAN	Leak from T200 or its outlet nozzle	- Corrosion of the tank floor, nozzle or flange - mechanical damage	Gas oil released into the tank area; a large leak cannot be stopped while the outlet is open.	- V200 hand valve (local only) - Operator rounds	4	2	8 M	R06	Process / Safety

Node N2 · Suction

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Node	N2 · Suction: V200, S-200 and the suction header
Design intent	Deliver liquid from T200 through V200 and strainer S-200 to the pump suctions with a small loss (3.5 m at 14 L/s), leaving NPSH available well above the NPSH required.	Design conditions	Suction head $\approx$ 0.5 m at the header; NPSH available 12.3 m vs required 4.5 m on the running pump (margin 7.8 m); 14.0 L/s
Equipment	T200 outlet line, V200 (gate), S-200 strainer, common suction header, V202 / V206	Instruments	PI-209 (local gauge only)
Drawing	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Document	PPS-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N2.1	Flow	NO	No flow to the pumps	- V200 closed in error - S-200 fully blinded	Running pump starves, cavitates and runs dry: seal faces overheat and leak, gas oil fire at the pump; feed to the cat cracker lost.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - TSHH-212 casing > 110 °C for 5 s $\rightarrow$ motor A tripped - Seal-leak alarm (gas detector at each pump) - Start permissives: SIS reset, T200 ≥ 15 %, no motor trip latched, suction valve open and casing vented (both confirmed in the field)	4	2	8 M	R07, R16	Electrical / Operations · Instrument
N2.2	Pressure	LESS	Suction pressure low (high suction loss)	- S-200 plugging with debris or scale (fault library: strainer plugging) - V200 or V202 left part-open	NPSH available falls below required: cavitation erosion, vibration, bearing and seal damage (release); flow falls toward 10 L/s.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - Local gauges PI-209 / PI-210 / PI-220 on operator rounds	3	3	9 M	R07, R08	Instrument · Instrument / Mechanical
N2.3	Pressure	LESS	NPSH available reduced by the liquid	- Light ends in the feed - feed hotter than 60 °C from the tank farm	Cavitation of the running pump with little change in flow, so it can go on unnoticed and damage impeller and seal.	Local vibration and noise on rounds	3	2	6 M	R03, R08	Instrument / Mechanical · Operations / Process
N2.4	Composition	AS WELL AS	Air or vapour drawn into the suction	- Vortex at a low T200 level - leak at the strainer cover gasket under suction	Gas pocket in the casing: loss of prime, dry running, seal failure.	- LI-200 LO 20 % / HI 85 % - LSLL-200 at 10 % for 5 s $\rightarrow$ both pumps tripped (SIS, first-out)	3	2	6 M	R16	Electrical / Operations
N2.5	Composition	PART OF	Strainer only partly effective	Basket torn or bypassed after cleaning	Debris reaches the impellers (fault library: impeller partly blocked): head and flow fall, vibration rises.	Strainer inspection at cleaning (procedure)	2	3	6 M	R13	Mechanical
N2.6	Measurement	NO	No remote suction measurement	- PI-209 is a local gauge - nothing in the control room shows the suction pressure	Strainer fouling or a throttled valve is found only when the flow is already short or the pump cavitates.	- Local gauges PI-209 / PI-210 / PI-220 on operator rounds - FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws)	2	4	8 M	R07	Instrument
N2.7	Pressure	MORE	Suction side pressurised from the discharge	Back-flow through a passing check valve and an idle pump with V200 closed (suction isolated)	Suction piping, strainer and V202/V206 see the discharge pressure (up to the shut-off head); a flange or the strainer cover can leak.	- Swing check valve NRV-210 / NRV-220 on each pump discharge - Permit to work with isolation of the equipment (written procedure)	3	2	6 M	R21, R32	Operations / Mechanical · Process / Mechanical

Node N3 · Pump A leg

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Node	N3 · Pump A leg (P-200A, normally running)
Design intent	P-200A (3600 rpm, 10 kW motor, shut-off 45 m) raises the gas oil from the suction header to about 32 m casing head at 14 L/s and delivers it through NRV-210 and V201 into the discharge header.	Design conditions	14.0 L/s, casing 32.4 m, 5.72 kW, 13.2 A (FLA 18.9 A), casing 60 °C, NPSH margin 7.8 m
Equipment	V202, P-200A and motor, vent VA, NRV-210, V201	Instruments	PI-210 (local), II-210, TI-212, TSHH-212, 49-210 overload relay
Drawing	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Document	PPS-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N3.1	Flow	NO	Running pump stops	- Motor A trips (protection relay, fault library) - power dip - single-phasing trips the overload	Feed to the cat cracker lost; D-150 empties and the cat cracker trips its charge pump; with the steam still on, the oil standing in E-100 heats toward the steam temperature.	- Motor-trip alarm in the DCS for each pump - FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - LI-150 LO 30 % / HI 80 % - LALL-150 at 10 % → cat-cracker charge pump tripped (cat-cracker side) - Spare pump lined up for a manual start	3	4	12 H	R09, R10	Instrument · Instrument / Operations
N3.2	Flow	NO	Pump running against a blocked discharge (deadhead)	- V201 closed in the field by mistake (fault library) - V100 closed by air failure, ESD or LSHH-150	All motor power goes into the casing liquid (about 6 °C per minute): seal faces flash, the seal fails, hot gas oil is released; prolonged deadhead can damage the casing.	- TI-212 HI 85 °C (casing A) - TSHH-212 casing > 110 °C for 5 s → motor A tripped - FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws)	4	3	12 H	R11, R12, R24	Instrument · Process / Mechanical
N3.3	Flow	LESS	Pump delivers less than its curve	- Wear rings worn (fault library) - impeller damage - V201 disc parted from its stem (fault library)	FRC-100 opens V100 to the limit and the feed is still short; at low flow the pump runs rough and hot.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - V100 output on the faceplate	2	4	8 M	R13, R25	Instrument · Mechanical
N3.4	Rotation	REVERSE	Pump A turns backwards	Two supply phases swapped after electrical work (fault library)	About 60 % of the head and far less flow; the impeller nut can work loose on reverse rotation and wreck the pump.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - Commissioning checks after electrical work (procedure)	3	3	9 M	R14	Electrical
N3.5	Current	MORE	Motor current high	- Single-phasing (fault library) - running out at high flow in parallel with V100 wide open - mechanical binding	Motor windings overheat: insulation damage, motor fire.	- II-210 HI 20 A - Overload relay 49-210: > 21.7 A for 10 s → motor A tripped	3	2	6 M	—	—
N3.6	Temperature	MORE	Casing hot from dry running	- Casing not vented before the start (air lock) - suction valve closed - loss of prime	Seal faces run dry and fail within about a minute (model): gas oil sprays from the seal, fire.	- Start permissives: SIS reset, T200 ≥ 15 %, no motor trip latched, suction valve open and casing vented (both confirmed in the field) - TSHH-212 casing > 110 °C for 5 s → motor A tripped - Seal-leak alarm (gas detector at each pump)	4	2	8 M	R16	Electrical / Operations

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N3.7	Pressure	MORE	Header pressure at shut-off	Discharge blocked (V100 or V201 closed) with the pump running	The header rises to the shut-off head plus the suction head; above the piping rating only if the rating was set for the operating point.	- Shut-off head A 45 m, B 46.5 m: the pumps cannot raise the header above shut-off plus the suction head - Piping rated for the shut-off head (to confirm)	2	2	4 L	R24	Instrument
N3.8	Integrity	OTHER THAN	Seal leak	Worn or damaged mechanical seal after cavitation, dry running or vibration	Gas oil at 60 °C sprays or pools under the pump; ignition on the hot motor or coupling gives a pool fire that can spread to the spare.	- Seal-leak alarm (gas detector at each pump) - ESD push-button: both pumps stopped, V100 and TV-110 closed	4	3	12 H	R06, R17	Mechanical · Process / Safety
N3.9	Measurement	NO	No remote view of the pump head	- PI-210 is a local gauge - II-210 is the only remote pump signal	Loss of head, wear or reverse rotation is seen late, from the flow and the valve position only.	- II-210 HI 20 A - Local gauges PI-209 / PI-210 / PI-220 on operator rounds	2	4	8 M	R18, R08	Instrument · Instrument / Mechanical

Node N4 · Pump B leg

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Node	N4 · Pump B leg (P-200B, spare) with the check valves
Design intent	P-200B (shut-off 46.5 m) stands by lined up — V206 and V205 open, casing vented and full — ready for a manual start; NRV-220 keeps the header pressure off the idle pump.	Design conditions	Idle: casing at the suction head (0.5 m), 53 °C, no flow; header 32.0 m held back by NRV-220
Equipment	V206, P-200B and motor, vent VB, NRV-220, V205	Instruments	PI-220 (local), II-220, TI-222, TSHH-222, 49-220 overload relay
Drawing	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Document	PPS-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N4.1	Flow	REVERSE	Back-flow through the idle pump	• NRV-220 flapper no longer seats (fault library) • check valve worn or held open by debris	Part of the running pump flow circulates back through the idle pump to the suction: the feed to the cat cracker is short with V100 wide open, D-150 falls, the idle pump is driven backwards.	• FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) • LI-150 LO 30 % / HI 80 % • Local gauges PI-209 / PI-210 / PI-220 on operator rounds	3	4	12 H	R18, R19, R20	Instrument · Instrument / Electrical · Mechanical
N4.2	Rotation	REVERSE	Spare started while turning backwards	• Back-flow spins the idle pump in reverse • the operator starts it as the spare	Motor starts against reverse rotation: long high-current start, coupling or shaft overstressed, seal damage and release.	• Overload relay 49-220: > 21.7 A for 10 s → motor B tripped • Procedure: start against a closed V205	3	3	9 M	R15, R19	Electrical / Mechanical · Instrument / Electrical
N4.3	Pressure	MORE	Isolated spare casing pressurised	• V206 closed for maintenance while V205 or NRV-220 passes • liquid blocked in and warmed by the sun	Casing and suction flange (rated for suction) see header pressure or thermal expansion: leak, or a release when the casing is opened.	• Permit to work with isolation of the equipment (written procedure) • Local gauges PI-209 / PI-210 / PI-220 on operator rounds	4	2	8 M	R21, R32	Operations / Mechanical · Process / Mechanical
N4.4	Flow	NO	Spare delivers nothing when needed	• Gas pocket in the casing after draining and refilling without venting (fault library: air lock) • V206 left closed	No flow from the spare; it runs dry and the seal fails; the feed stays lost and the cat cracker trips.	• Start permissives: SIS reset, T200 ≥ 15 %, no motor trip latched, suction valve open and casing vented (both confirmed in the field) • TSHH-222 casing > 110 °C for 5 s → motor B tripped • Seal-leak alarm (gas detector at each pump)	4	2	8 M	R16, R09	Electrical / Operations · Instrument / Operations
N4.5	Flow	LESS	Weak pump pushed back in parallel	• Impeller partly blocked (fault library) • two unequal pumps in parallel at low demand (shut-off A 45.0 m vs B 46.5 m)	The weaker pump runs near shut-off: it heats, vibrates, and with a passing check valve can even be pushed backwards.	• TI-222 HI 85 °C (casing B) • TSHH-222 casing > 110 °C for 5 s → motor B tripped • II-220 HI 20 A	3	3	9 M	R11, R12	Instrument · Process / Mechanical
N4.6	Temperature	MORE	Idle casing heats	• Back-flow through the idle casing combined with a closed suction valve • sun on a blocked-in casing	Trapped liquid expands: flange or seal leak.	TI-222 HI 85 °C (casing B)	2	2	4 L	R21	Process / Mechanical

Node N5 · Discharge header, FE-100, V100 and FRC-100

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Node	N5 · Discharge header, FE-100, V100 and FRC-100
Design intent	Carry the pump discharge (about 32 m) through FE-100 and V100 (modified equal percentage, fail closed) at the 14 L/s asked for by the cat cracker; FRC-100 (PI, remote set point) holds the flow with V100 near 84 %.	Design conditions	14.0 L/s at 32.0 m header head; V100 84 %; FRC-100 Kc 0.35, Ti 3 s, span 0–20 L/s
Equipment	Common discharge header, low-point drain DRN, FE-100, V100	Instruments	FRC-100 (FT-100), V100 positioner
Drawing	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Document	PPS-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N5.1	Flow	NO	No flow through V100	- Instrument-air failure (fault library), ESD or LSHH-150 closes V100 - V100 sticks closed	Running pump deadheaded (see N3) and the cat cracker loses its feed; D-150 empties.	- V100 fails closed (instrument air, ESD, LSHH-150) - TI-212 HI 85 °C (casing A) - TSHH-212 casing > 110 °C for 5 s → motor A tripped - FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - LI-150 LO 30 % / HI 80 %	4	3	12 H	R11, R12	Instrument · Process / Mechanical
N5.2	Flow	LESS	Feed short of the set point	- Check valve on the idle pump passing (N4) - V100 sticks while the cat cracker raises its rate (fault library) - pump wear - strainer fouling	D-150 falls; at LALL-150 the cat cracker loses its charge pump and goes to hot standby (large production loss, a hazardous restart).	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - LI-150 LO 30 % / HI 80 % - LALL-150 at 10 % → cat-cracker charge pump tripped (cat-cracker side)	3	3	9 M	R25, R09	Instrument · Instrument / Operations
N5.3	Flow	MORE	Real flow higher than measured	FT-100 reads low (fault library: impulse line / transmitter drift)	FRC-100 opens V100 and sends more than the cat cracker draws: D-150 rises; colder feed out of E-100.	- LI-150 LO 30 % / HI 80 % - LSHH-150 at 95 % → V100 forced closed (latched until reset)	2	3	6 M	R26	Instrument
N5.4	Flow	MORE	Set point too high	- Remote set point from the cat cracker wrong - FRC-100 switched to a local set point in error	More feed than charge: D-150 fills; pump runs out on its curve with more power and less NPSH margin.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - LI-150 LO 30 % / HI 80 % - LSHH-150 at 95 % → V100 forced closed (latched until reset)	2	3	6 M	R25	Instrument
N5.5	Flow	REVERSE	D-150 drains back into the header	Both pumps stop (power dip, LSL-200) with V100 open and a passing check valve	Hot feed flows back from D-150 (1.5 bar g, 6 m up) through E-100, V100 and the idle pump into T200: pumps spin backwards, D-150 empties, oil above 100 °C reaches a tank that may hold water.	- Swing check valve NRV-210 / NRV-220 on each pump discharge - LALL-150 at 10 % → cat-cracker charge pump tripped (cat-cracker side)	4	2	8 M	R22, R20	Instrument · Mechanical
N5.6	Flow	AS WELL AS	Flow to the drain as well	Header low-point drain left open after maintenance (fault library)	Gas oil runs to the oily-water sewer (about 3 L/s): oil in the drains, fire risk in the sewer, feed short.	FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws)	3	3	9 M	R23, R26	Instrument · Operations
N5.7	Flow	PART OF	V100 does not close fully	Seat leakage of V100 when LSHH-150 or ESD forces it closed	D-150 keeps filling slowly after the high-high trip.	LI-150 LO 30 % / HI 80 %	3	2	6 M	R30	Instrument

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N5.8	Pressure	MORE	Blocked-in header heats up	Header blocked in between the check valves and a closed V100 (shutdown) and warmed by the sun or a fire	Thermal expansion of trapped liquid: very high pressure, flange or gasket leak.	None specific	3	2	6 M	R21	Process / Mechanical
N5.9	Control	OTHER THAN	FRC-100 not in its normal mode	Left in MAN or on a local set point after start-up	The feed no longer follows the cat cracker: D-150 drifts to a level alarm.	- LI-150 LO 30 % / HI 80 % - FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws)	2	3	6 M	—	—

Node N6 · Preheater E-100

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Node	N6 · Preheater E-100: steam side and feed side
Design intent	Heat the feed from 60 to 150 °C (about 2.3 MW) with steam supplied at 11 bar a through TV-110 (fail closed; TIC-110 normally in MAN near 50 %); shell near 6 bar a, condensate through a trap.	Design conditions	2.31 MW, steam 4.03 t/h, shell 5.97 bar a (159 °C), outlet 150 °C, TV-110 50 %
Equipment	E-100 (steam in the shell), TV-110, steam trap, condensate return	Instruments	TIC-110 (TI-110), PI-111, FI-112
Drawing	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Document	PPS-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N6.1	Temperature	MORE	Feed outlet too hot	TV-110 driven open (TIC-110 output to 100 % or a set-point error)	Hotter feed to D-150 (light components flash); feed preheat upset for the cat cracker.	TI-110 LO 140 °C (with flow) / HI 165 °C	2	3	6 M	R29	Instrument
N6.2	Temperature	MORE	Oil heated with no flow	Feed flow lost (pump trip, V100 closed, cat cracker cut) while the steam stays on	The oil standing in the tubes heats toward the steam temperature (184 °C at the supply pressure): coking and fouling of the tubes; a slug of overheated oil at the restart.	- TI-110 LO 140 °C (with flow) / HI 165 °C - ESD push-button: both pumps stopped, V100 and TV-110 closed	3	4	12 H	R10	Instrument
N6.3	Temperature	LESS	Feed outlet too cold	- Steam supply pressure low (fault library) - TV-110 fails closed - E-100 fouling (fault library) - trap stuck closed and the shell floods	Cold feed to the cat cracker: lower conversion, more coke (operability).	- TI-110 LO 140 °C (with flow) / HI 165 °C - PI-111 and FI-112 indication	2	3	6 M	R29	Instrument
N6.4	Pressure	MORE	Shell at the steam supply pressure	TV-110 wide open with little condensing load (no feed flow)	The shell reaches the supply pressure (11 bar a); if its design pressure was set for the operating point, the shell is overpressured.	- TV-110 fails closed (instrument air, ESD) - Shell design pressure (to confirm)	4	2	8 M	R27	Process
N6.5	Integrity	OTHER THAN	Tube leak or rupture	Corrosion, thermal fatigue	Steam enters the oil (water to D-150 and the cat cracker) or oil enters the shell and the condensate system.	None specific	3	2	6 M	R27, R28	Instrument / Mechanical · Process
N6.6	Level	MORE	Condensate builds up in the shell	Steam trap fails closed	Shell floods, duty falls; water hammer when the trap clears can break the tube bundle or a flange.	TI-110 LO 140 °C (with flow) / HI 165 °C	3	2	6 M	R28	Instrument / Mechanical
N6.7	Measurement	OTHER THAN	TI-110 reads wrong	Thermocouple reads low (fault library)	The operator raises the steam to the false reading: real outlet too hot; with TIC-110 in AUTO the controller does it automatically.	TI-110 LO 140 °C (with flow) / HI 165 °C	2	3	6 M	R29	Instrument

Node N7 · D-150 and the cat-cracker interface

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Node	N7 · D-150 and the cat-cracker interface
Design intent	Hold up to 30 m ³ of hot feed (design 50 %, 1.5 bar g) as surge between the station and the cat cracker, which draws 14 L/s (FI-151); the station does not control the D-150 level — the cat cracker resets FRC-100 remotely.	Design conditions	50 % = 15 m ³ ; 14.3 min to LALL-150 with no feed, 16.1 min to LSHH-150 with no charge (at 14 L/s)
Equipment	D-150, charge line to the cat cracker	Instruments	LI-150, LSHH-150, LALL-150 (cat-cracker side), FI-151
Drawing	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Document	PPS-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N7.1	Level	MORE	D-150 fills	- Cat-cracker charge pump trips or the cat cracker cuts its feed (fault library) while the station keeps feeding - FT-100 reads low	D-150 overfills: hot oil to the cat-cracker vent or relief, liquid carry-over; if V100 is then closed, the running pump is deadheaded.	- LI-150 LO 30 % / HI 80 % - LSHH-150 at 95 % → V100 forced closed (latched until reset) - TSHH-212 casing > 110 °C for 5 s → motor A tripped	4	3	12 H	R12, R30	Instrument
N7.2	Level	LESS	D-150 runs down	- Feed short or lost (any cause in N1-N5) - the cat cracker raises its charge (fault library)	LALL-150 trips the cat-cracker charge pump: the cat cracker goes to hot standby (large production loss, a hazardous transient).	- LI-150 LO 30 % / HI 80 % - LALL-150 at 10 % → cat-cracker charge pump tripped (cat-cracker side) - Spare pump lined up for a manual start	3	3	9 M	R09, R26	Instrument · Instrument / Operations
N7.3	Pressure	MORE	D-150 overpressure	- Water or light ends flash in the hot drum - outlet blocked by the cat cracker	Drum overpressure; relief to the cat-cracker flare.	D-150 relief on the cat-cracker unit (not on PPS-PID-001)	4	2	8 M	R31, R04	Operations · Process
N7.4	Flow	MORE	Charge drawn above the feed	Cat cracker raises its charge without resetting FRC-100 (fault library)	D-150 falls slowly to the low-level alarm and LALL-150.	- LI-150 LO 30 % / HI 80 % - FI-151 indication	2	4	8 M	R26	Instrument
N7.5	Measurement	OTHER THAN	LI-150 reads wrong	- Level transmitter sticks or drifts - LI-150 alarm and LSHH-150 trip read one measurement	High-level alarm and trip both blind: D-150 overfills.	FI-151 vs FRC-100 seen by the operator	4	2	8 M	R30	Instrument
N7.6	Control	OTHER THAN	Level not controlled by the station	No level loop: D-150 is balanced only by the cat cracker resetting FRC-100	Any mismatch between the remote set point and the real charge drifts the level until an alarm.	LI-150 LO 30 % / HI 80 %	2	3	6 M	R26	Instrument

Node N8 · Procedures

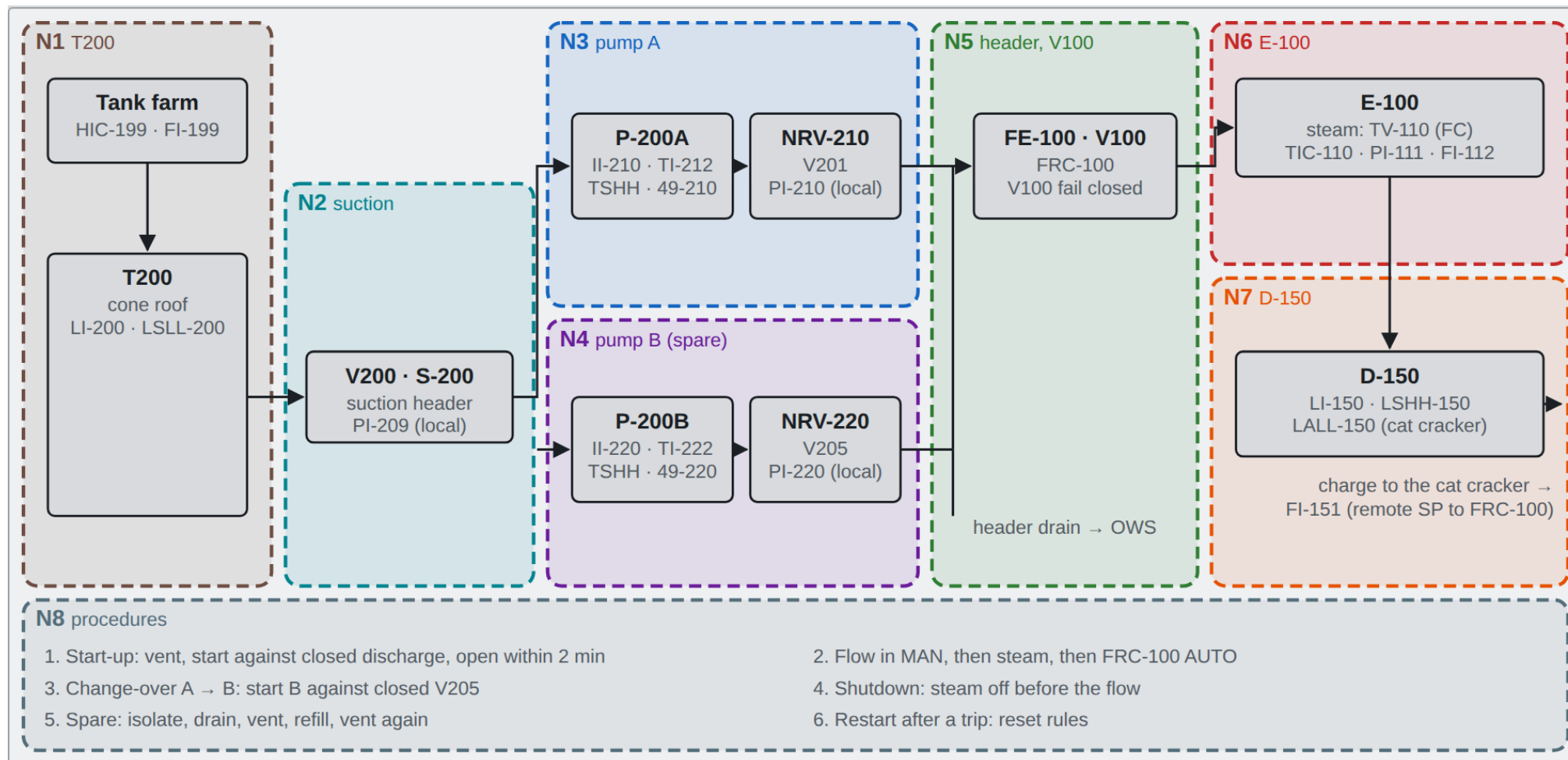
Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Node	N8 · Procedures: start-up, change-over, shutdown, maintenance of the spare
Design intent	Start a pump vented, against a closed discharge, open it within 2 min; establish flow before steam; change over A → B by starting B against a closed V205; shut down steam before flow; isolate, drain and vent the spare safely for maintenance.	Design conditions	Shut-off check ≥ 40 m on the local gauge before opening the discharge; deadhead heating about 6 °C/min; 14.3 min of D-150 hold-up from 50 % to LALL-150
Equipment	P-200A/B, V201/V205, V202/V206, vents VA/VB, V100, TV-110	Instruments	PI-210/220 (local), FRC-100, TIC-110, II-210/220, TI-212/222
Drawing	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Document	PPS-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N8.1	Sequence	BEFORE	Steam on before the feed flows	Start-up order not followed: TV-110 opened before V100	Oil standing in E-100 heats toward the steam temperature; the first oil to D-150 is too hot.	- Written procedures: start-up, change-over A → B, shutdown (Exercises tab) - TI-110 LO 140 °C (with flow) / HI 165 °C	2	3	6 M	R10	Instrument
N8.2	Sequence	BEFORE	Pump started before it is vented or with the suction closed	- Venting skipped - suction valve not opened (fault library: air lock)	Dry running: the seal fails in about a minute and sprays gas oil; fire.	- Start permissives: SIS reset, T200 ≥ 15 %, no motor trip latched, suction valve open and casing vented (both confirmed in the field) - Seal-leak alarm (gas detector at each pump)	4	3	12 H	R16	Electrical / Operations
N8.3	Timing	LATE	Discharge opened late after a start against the closed valve	- Operator distracted after the start - V201 hard to open	Casing heats at shut-off (about 6 °C per minute): seal damage, motor trip.	- TI-212 HI 85 °C (casing A) - TSHH-212 casing > 110 °C for 5 s → motor A tripped - Procedure: open within 2 min	3	3	9 M	R11	Process / Mechanical
N8.4	Timing	LATE	Spare not started in time after a trip	Trip not noticed, spare not ready, operator busy elsewhere	D-150 runs down and LALL-150 trips the cat cracker.	- Motor-trip alarm in the DCS for each pump - FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - LI-150 LO 30 % / HI 80 %	3	4	12 H	R09, R33	Instrument / Operations · Operations / Training
N8.5	Timing	EARLY	Cat cracker draws before the feed is established	Cat cracker told the feed is ready before FRC-100 is in AUTO at rate	D-150 runs down during the start-up; LALL-150 trips the charge pump.	- LI-150 LO 30 % / HI 80 % - LALL-150 at 10 % → cat-cracker charge pump tripped (cat-cracker side)	2	3	6 M	—	—
N8.6	Sequence	AFTER	Steam closed after the flow is stopped	Shutdown order reversed: V100 closed first, TV-110 later	Oil blocked in E-100 heats toward the steam temperature: coking, thermal expansion against the closed V100.	- Written procedures: start-up, change-over A → B, shutdown (Exercises tab) - TI-110 LO 140 °C (with flow) / HI 165 °C	3	3	9 M	R10, R21	Instrument · Process / Mechanical
N8.7	Sequence	OTHER THAN	Wrong pump or wrong valve operated	- Running pump stopped before the spare delivers - spare started against an open V205 while NRV-220 passes - wrong valve tag closed in the field (fault library: V201 closed)	Feed dip or loss; start against reverse rotation; deadhead of the running pump.	- Written procedures: start-up, change-over A → B, shutdown (Exercises tab) - FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - TSHH-212 casing > 110 °C for 5 s → motor A tripped	3	3	9 M	R15, R19, R33	Electrical / Mechanical · Instrument / Electrical · Operations / Training

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N8.8	Maintenance	OTHER THAN	Spare opened while still under pressure	- Isolation by single valves that pass - casing not drained and vented before the seal is changed	Gas oil at about 60 °C and header pressure released onto the fitter: injury, fire.	- Permit to work with isolation of the equipment (written procedure) - Local gauges PI-209 / PI-210 / PI-220 on operator rounds	4	3	12 H	R32, R21	Operations / Mechanical · Process / Mechanical
N8.9	Maintenance	AS WELL AS	Casing drained to grade	Drained into an open tray or onto the ground instead of a closed drain	Gas oil spill and vapour in the pump area; fire risk.	Permit to work with isolation of the equipment (written procedure)	3	3	9 M	R32, R17	Mechanical · Operations / Mechanical
N8.10	Maintenance	PART OF	Casing refilled but not fully vented	Vent closed when the first liquid shows, gas still trapped	Air lock on the next start: no flow, dry running, seal failure.	Start permissives: SIS reset, T200 ≥ 15 %, no motor trip latched, suction valve open and casing vented (both confirmed in the field)	3	3	9 M	R16	Electrical / Operations
N8.11	Electrical work	PART OF	Motor returned to service without a rotation check	Electrical work closed out without a bump test (fault library: phases swapped)	Pump runs backwards at the next start: low flow, possible impeller damage.	FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws)	3	3	9 M	R14	Electrical
N8.12	Restart	BEFORE	Restart before the cause of a trip is cleared	Reset pressed with the level just above the permissive or a hot casing	Repeat trip or further damage to a hot pump.	Reset rules: refused while T200 < 15 %, a casing > 85 °C, a protection relay still tripped or D-150 > 90 %	2	2	4 L	—	—

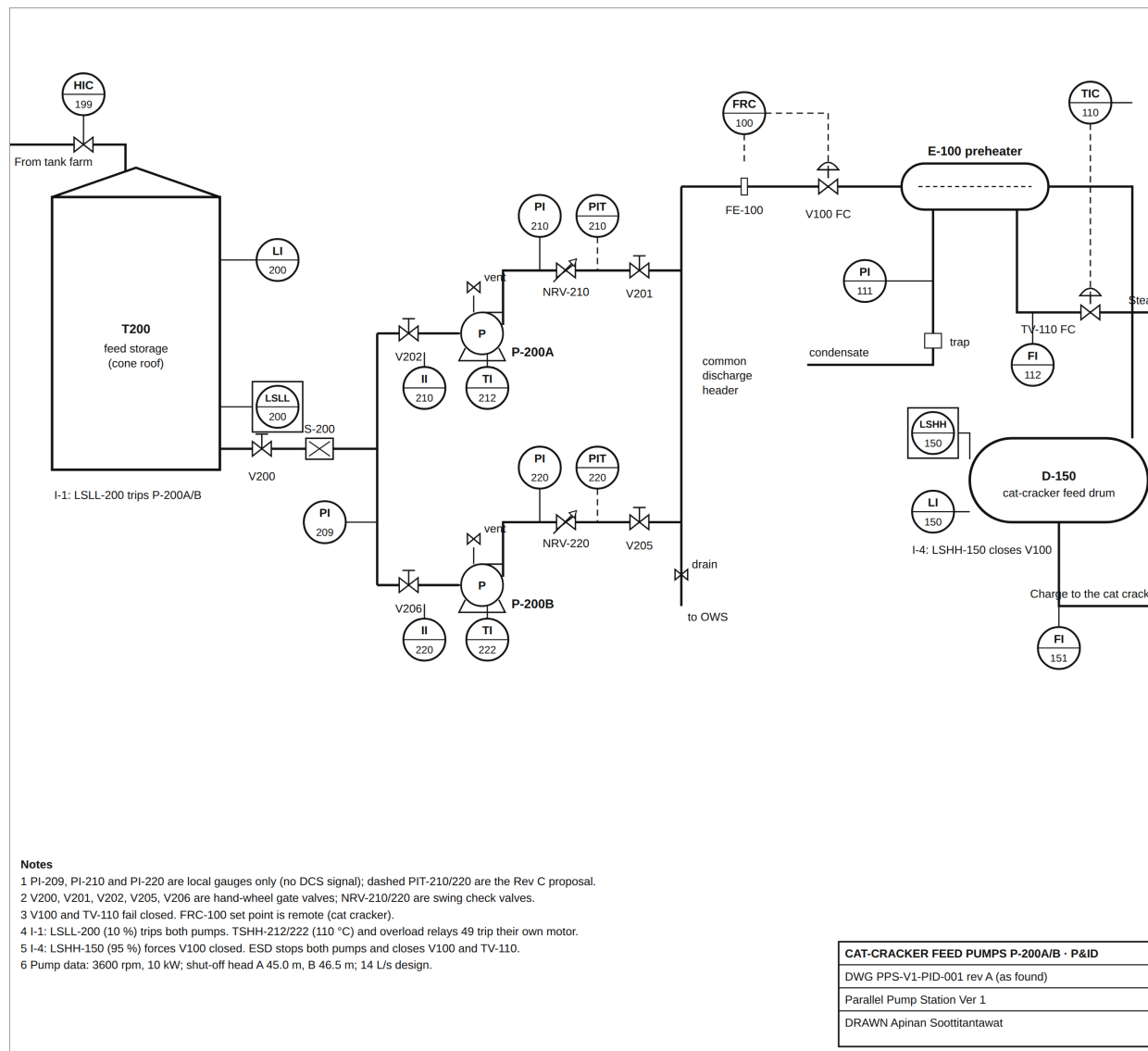
Appendix 2 · HAZOP documents (drawings)

A2-1 · HAZOP node map



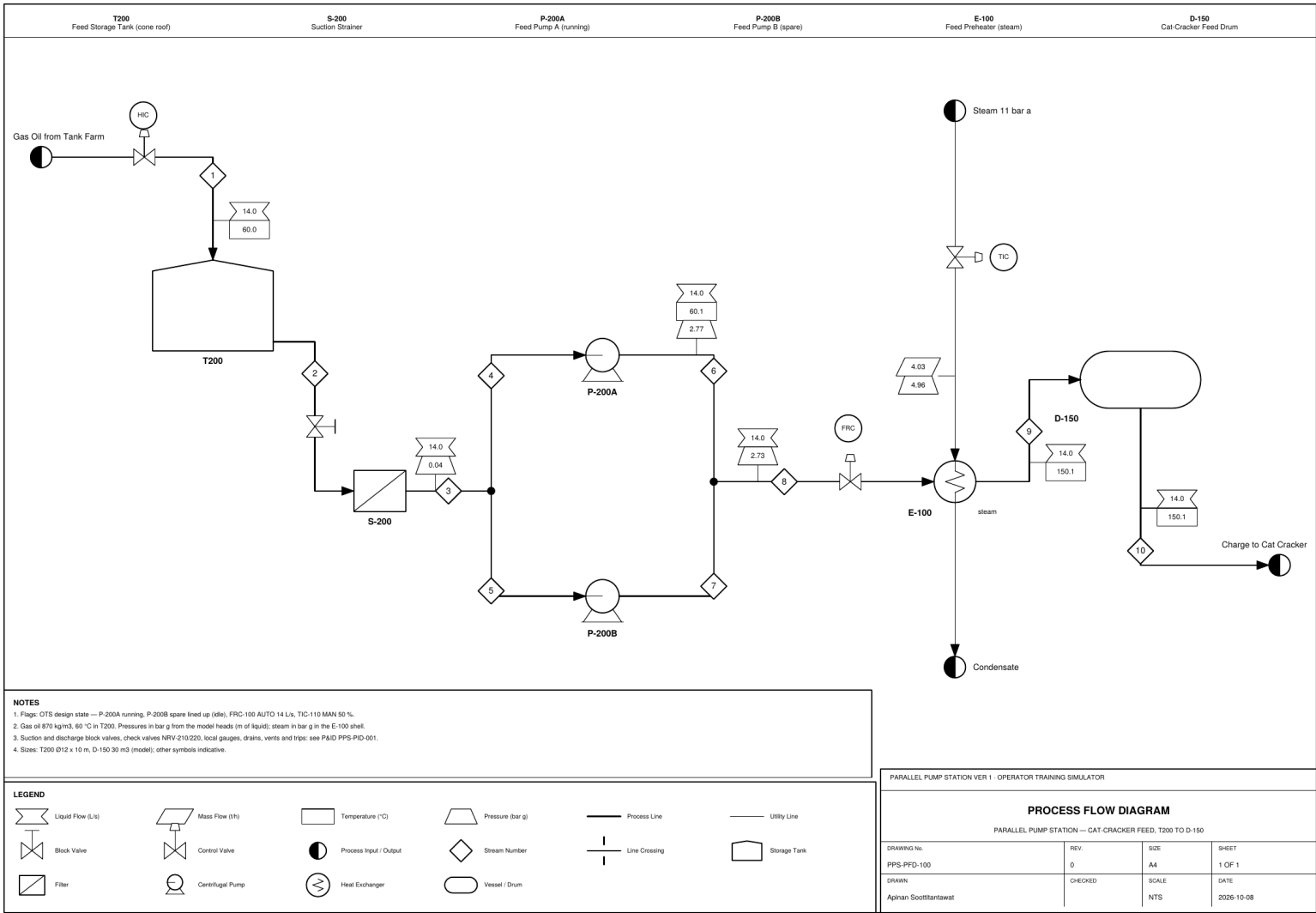
A2-1. HAZOP node map of the pump station (nodes N1–N7; N8 = procedures).

A2-3 · PPS-V1-PID-001 (screen P&ID of the DCS console and the exercises)



A2-3. The same unit drawn in the DCS layout used on the console (P&ID tab, below the EPC sheet).

A2-4 · PPS-PFD-100 (process flow diagram)



A2-4. Process flow diagram PPS-PFD-100 with the stream numbers of the Process tab (PFD tab).

Appendix 3 · HAZOP action close-out sheets

One sheet per recommendation (33 sheets). The action party writes the response, ticks the status and signs; the HAZOP leader accepts the close-out.

HAZOP ACTION CLOSE-OUT SHEET · R01

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R01
Node	N1 · Tank-farm transfer and T200	Action party	Instrument
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.1	Flow NO : No transfer into T200	- Tank-farm transfer pump stops (fault library: transfer lost) - HIC-199 set to 0 or its valve fails closed - block valve at the tank farm closed	T200 runs down while the pumps draw; at low level the pumps lose suction (vortex, air), run dry, the seals fail and hot gas oil is released; the cat cracker loses its feed.	- LI-200 LO 20 % / HI 85 % - LSLL-200 at 10 % for 5 s → both pumps tripped (SIS, first-out) - FI-199 indication - Hold-up: hours of feed at design	S3 L2 6 Medium
N1.3	Level LESS : Level low with the instruments blind	- LI-200 transmitter sticks or reads high - one measurement serves the LI-200 alarm, LSL-200 and LSLL-200	No alarm and no trip while the real level falls: both pumps run dry, seals fail, gas oil is released at the pumps (fire).	- FI-199 vs FRC-100 mismatch seen by the operator - Field level gauge on rounds (assumed)	S4 L2 8 Medium

HAZOP recommendation

Give LSLL-200 its own level transmitter LT-200B (today the LI-200 alarm, the LSL-200 start permissive and the LSLL-200 trip read one measurement).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R02

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R02
Node	N1 · Tank-farm transfer and T200	Action party	Instrument / Operations
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.2	Flow MORE : Transfer larger than the draw	- HIC-199 left at 100 % (20 L/s) when the feed rate falls - the cat cracker trips and the station is cut back	Level rises until T200 overflows through the roof vents: gas oil pool in the tank area, fire if ignited.	- LI-200 LO 20 % / HI 85 % - Operator rounds at the tank	S4 L2 8 Medium
N1.4	Level MORE : Overfill of T200	- LI-200 reads low - transfer left on during a feed outage	Overflow of gas oil at about 60 °C into the tank area; vapour near the flash point; pool fire.	- LI-200 LO 20 % / HI 85 % - Tank-farm operator watches the transfer	S4 L2 8 Medium

HAZOP recommendation

Independent high-high level LSHH-200 on T200 that stops the tank-farm transfer (overfill protection to API 2350).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R03

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R03
Node	N1/N2 · Tank-farm transfer and T200 / Suction	Action party	Operations / Process
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.5	Composition AS WELL AS : Light ends (naphtha) in the feed	- Wrong tank lined up at the tank farm - light material left in the line (fault library: light ends in the feed)	Vapour pressure rises toward 75 kPa, NPSH available falls by about 8 m: the pumps cavitate (erosion, vibration, seal damage); flammable vapour in the T200 roof space.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - Local vibration and noise at the pumps on rounds	S4 L2 8 Medium
N2.3	Pressure LESS : NPSH available reduced by the liquid	- Light ends in the feed - feed hotter than 60 °C from the tank farm	Cavitation of the running pump with little change in flow, so it can go on unnoticed and damage impeller and seal.	Local vibration and noise on rounds	S3 L2 6 Medium

HAZOP recommendation

Feed quality at the tank farm: vapour-pressure / flash-point check of each transfer and a line-up check, so naphtha or light material cannot reach T200 unseen.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R04

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R04
Node	N1/N7 · Tank-farm transfer and T200 / D-150 and the cat-cracker interface	Action party	Operations
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.6	Composition AS WELL AS: Water in the feed	Water settled in the tank-farm tank or in T200 drawn with the gas oil	Water heated in E-100 flashes in D-150 (water boils at about 128 °C at 1.5 bar g): pressure surge in D-150 and steam carried to the cat cracker.	T200 water draw-off by procedure (assumed)	S4 L2 8 Medium
N7.3	Pressure MORE: D-150 overpressure	- Water or light ends flash in the hot drum - outlet blocked by the cat cracker	Drum overpressure; relief to the cat-cracker flare.	D-150 relief on the cat-cracker unit (not on PPS-PID-001)	S4 L2 8 Medium

HAZOP recommendation

T200 water draw-off routine (each shift) and water detection at the tank outlet before the feed is heated in E-100.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R05

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R05
Node	N1 · Tank-farm transfer and T200	Action party	Mechanical
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.7	Pressure LESS : Vacuum in T200	P/V vent or flame arrester blocked while the pumps draw 14–20 L/s	Cone-roof tank pulled under vacuum: roof or shell buckles, the tank leaks.	Tank vent (not shown on PPS-PID-001)	S3 L2 6 Medium

HAZOP recommendation

Confirm the T200 pressure/vacuum vent (API 2000) for the largest pump-out and fill rates; inspect the flame arrester.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R06

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R06
Node	N1/N3 · Tank-farm transfer and T200 / Pump A leg	Action party	Process / Safety
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.8	Integrity OTHER THAN: Leak from T200 or its outlet nozzle	- Corrosion of the tank floor, nozzle or flange - mechanical damage	Gas oil released into the tank area; a large leak cannot be stopped while the outlet is open.	- V200 hand valve (local only) - Operator rounds	S4 L2 8 Medium
N3.8	Integrity OTHER THAN: Seal leak	Worn or damaged mechanical seal after cavitation, dry running or vibration	Gas oil at 60 °C sprays or pools under the pump; ignition on the hot motor or coupling gives a pool fire that can spread to the spare.	- Seal-leak alarm (gas detector at each pump) - ESD push-button: both pumps stopped, V100 and TV-110 closed	S4 L3 12 High

HAZOP recommendation

Remotely operated, fire-safe emergency isolation valve EIV-200 on the T200 outlet, closable from the control room.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R07

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R07
Node	N2 · Suction	Action party	Instrument
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.1	Flow NO : No flow to the pumps	• V200 closed in error • S-200 fully blinded	Running pump starves, cavitates and runs dry: seal faces overheat and leak, gas oil fire at the pump; feed to the cat cracker lost.	• FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) • TSHH-212 casing > 110 °C for 5 s → motor A tripped • Seal-leak alarm (gas detector at each pump) • Start permissives: SIS reset, T200 ≥ 15 %, no motor trip latched, suction valve open and casing vented (both confirmed in the field)	S4 L2 8 Medium
N2.2	Pressure LESS : Suction pressure low (high suction loss)	• S-200 plugging with debris or scale (fault library: strainer plugging) • V200 or V202 left part-open	NPSH available falls below required: cavitation erosion, vibration, bearing and seal damage (release); flow falls toward 10 L/s.	• FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) • Local gauges PI-209 / PI-210 / PI-220 on operator rounds	S3 L3 9 Medium
N2.6	Measurement NO : No remote suction measurement	• PI-209 is a local gauge • nothing in the control room shows the suction pressure	Strainer fouling or a throttled valve is found only when the flow is already short or the pump cavitates.	• Local gauges PI-209 / PI-210 / PI-220 on operator rounds • FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws)	S2 L4 8 Medium

HAZOP recommendation

Suction pressure transmitter PIT-209 with a low alarm in the DCS, and a differential-pressure gauge across S-200.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R08

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R08
Node	N2/N3 · Suction / Pump A leg	Action party	Instrument / Mechanical
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.5	Composition AS WELL AS : Light ends (naphtha) in the feed	- Wrong tank lined up at the tank farm - light material left in the line (fault library: light ends in the feed)	Vapour pressure rises toward 75 kPa, NPSH available falls by about 8 m: the pumps cavitate (erosion, vibration, seal damage); flammable vapour in the T200 roof space.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - Local vibration and noise at the pumps on rounds	S4 L2 8 Medium
N2.2	Pressure LESS : Suction pressure low (high suction loss)	- S-200 plugging with debris or scale (fault library: strainer plugging) - V200 or V202 left part-open	NPSH available falls below required: cavitation erosion, vibration, bearing and seal damage (release); flow falls toward 10 L/s.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - Local gauges PI-209 / PI-210 / PI-220 on operator rounds	S3 L3 9 Medium
N2.3	Pressure LESS : NPSH available reduced by the liquid	- Light ends in the feed - feed hotter than 60 °C from the tank farm	Cavitation of the running pump with little change in flow, so it can go on unnoticed and damage impeller and seal.	Local vibration and noise on rounds	S3 L2 6 Medium
N3.9	Measurement NO : No remote view of the pump head	- PI-210 is a local gauge - II-210 is the only remote pump signal	Loss of head, wear or reverse rotation is seen late, from the flow and the valve position only.	- II-210 HI 20 A - Local gauges PI-209 / PI-210 / PI-220 on operator rounds	S2 L4 8 Medium

HAZOP recommendation

Vibration transmitters on P-200A/B with DCS alarms (VI-210/220 are local today).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R09

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R09
Node	N3/N7/N8 · Pump A leg / D-150 and the cat-cracker interface / Procedures	Action party	Instrument / Operations
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.1	Flow NO : Running pump stops	- Motor A trips (protection relay, fault library) - power dip - single-phasing trips the overload	Feed to the cat cracker lost; D-150 empties and the cat cracker trips its charge pump; with the steam still on, the oil standing in E-100 heats toward the steam temperature.	- Motor-trip alarm in the DCS for each pump - FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - LI-150 LO 30 % / HI 80 % - LALL-150 at 10 % → cat-cracker charge pump tripped (cat-cracker side) - Spare pump lined up for a manual start	S3 L4 12 High
N4.4	Flow NO : Spare delivers nothing when needed	- Gas pocket in the casing after draining and refilling without venting (fault library: air lock) - V206 left closed	No flow from the spare; it runs dry and the seal fails; the feed stays lost and the cat cracker trips.	- Start permissives: SIS reset, T200 ≥ 15 %, no motor trip latched, suction valve open and casing vented (both confirmed in the field) - TSHH-222 casing > 110 °C for 5 s → motor B tripped - Seal-leak alarm (gas detector at each pump)	S4 L2 8 Medium
N5.2	Flow LESS : Feed short of the set point	- Check valve on the idle pump passing (N4) - V100 sticks while the cat cracker raises its rate (fault library) - pump wear - strainer fouling	D-150 falls; at LALL-150 the cat cracker loses its charge pump and goes to hot standby (large production loss, a hazardous restart).	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - LI-150 LO 30 % / HI 80 % - LALL-150 at 10 % → cat-cracker charge pump tripped (cat-cracker side)	S3 L3 9 Medium
N7.2	Level LESS : D-150 runs down	- Feed short or lost (any cause in N1-N5) - the cat cracker raises its charge (fault library)	LALL-150 trips the cat-cracker charge pump; the cat cracker goes to hot standby (large production loss, a hazardous transient).	- LI-150 LO 30 % / HI 80 % - LALL-150 at 10 % → cat-cracker charge pump tripped (cat-cracker side) - Spare pump lined up for a manual start	S3 L3 9 Medium
N8.4	Timing LATE : Spare not started in time after a trip	Trip not noticed, spare not ready, operator busy elsewhere	D-150 runs down and LALL-150 trips the cat cracker.	- Motor-trip alarm in the DCS for each pump - FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - LI-150 LO 30 % / HI 80 %	S3 L4 12 High

HAZOP recommendation

Auto-start of the spare pump on loss of the running pump (motor trip or low header pressure), with the zero-speed permissive of R15.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R10

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R10
Node	N3/N6/N8 · Pump A leg / Preheater E-100 / Procedures	Action party	Instrument
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.1	Flow NO : Running pump stops	• Motor A trips (protection relay, fault library) • power dip • single-phasing trips the overload	Feed to the cat cracker lost; D-150 empties and the cat cracker trips its charge pump; with the steam still on, the oil standing in E-100 heats toward the steam temperature.	• Motor-trip alarm in the DCS for each pump • FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) • LI-150 LO 30 % / HI 80 % • LALL-150 at 10 % → cat-cracker charge pump tripped (cat-cracker side) • Spare pump lined up for a manual start	S3 L4 12 High
N6.2	Temperature MORE : Oil heated with no flow	Feed flow lost (pump trip, V100 closed, cat cracker cut) while the steam stays on	The oil standing in the tubes heats toward the steam temperature (184 °C at the supply pressure): coking and fouling of the tubes; a slug of overheated oil at the restart.	• TI-110 LO 140 °C (with flow) / HI 165 °C • ESD push-button: both pumps stopped, V100 and TV-110 closed	S3 L4 12 High
N8.1	Sequence BEFORE : Steam on before the feed flows	Start-up order not followed: TV-110 opened before V100	Oil standing in E-100 heats toward the steam temperature; the first oil to D-150 is too hot.	• Written procedures: start-up, change-over A → B, shutdown (Exercises tab) • TI-110 LO 140 °C (with flow) / HI 165 °C	S2 L3 6 Medium
N8.6	Sequence AFTER : Steam closed after the flow is stopped	Shutdown order reversed: V100 closed first, TV-110 later	Oil blocked in E-100 heats toward the steam temperature: coking, thermal expansion against the closed V100.	• Written procedures: start-up, change-over A → B, shutdown (Exercises tab) • TI-110 LO 140 °C (with flow) / HI 165 °C	S3 L3 9 Medium

HAZOP recommendation

Steam cut-off: close TV-110 automatically when the feed flow is lost (FT-100 low or both pumps stopped).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R11

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R11
Node	N3/N5/N8 · Pump A leg / Discharge header, FE-100, V100 and FRC-100 / Procedures	Action party	Process / Mechanical
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.2	Flow NO : Pump running against a blocked discharge (deadhead)	• V201 closed in the field by mistake (fault library) • V100 closed by air failure, ESD or LSHH-150	All motor power goes into the casing liquid (about 6 °C per minute): seal faces flash, the seal fails, hot gas oil is released; prolonged deadhead can damage the casing.	• TI-212 HI 85 °C (casing A) • TSHH-212 casing > 110 °C for 5 s → motor A tripped • FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws)	S4 L3 12 High
N4.5	Flow LESS : Weak pump pushed back in parallel	• Impeller partly blocked (fault library) • two unequal pumps in parallel at low demand (shut-off A 45.0 m vs B 46.5 m)	The weaker pump runs near shut-off: it heats, vibrates, and with a passing check valve can even be pushed backwards.	• TI-222 HI 85 °C (casing B) • TSHH-222 casing > 110 °C for 5 s → motor B tripped • II-220 HI 20 A	S3 L3 9 Medium
N5.1	Flow NO : No flow through V100	• Instrument-air failure (fault library), ESD or LSHH-150 closes V100 • V100 sticks closed	Running pump deadheaded (see N3) and the cat cracker loses its feed; D-150 empties.	• V100 fails closed (instrument air, ESD, LSHH-150) • TI-212 HI 85 °C (casing A) • TSHH-212 casing > 110 °C for 5 s → motor A tripped • FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) • LI-150 LO 30 % / HI 80 %	S4 L3 12 High
N8.3	Timing LATE : Discharge opened late after a start against the closed valve	• Operator distracted after the start • V201 hard to open	Casing heats at shut-off (about 6 °C per minute): seal damage, motor trip.	• TI-212 HI 85 °C (casing A) • TSHH-212 casing > 110 °C for 5 s → motor A tripped • Procedure: open within 2 min	S3 L3 9 Medium

HAZOP recommendation

Minimum-flow recycle from each pump discharge (upstream of the check valve) back to T200, sized for the API 610 minimum continuous flow.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R12

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R12
Node	N3/N5/N7 · Pump A leg / Discharge header, FE-100, V100 and FRC-100 / D-150 and the cat-cracker interface	Action party	Instrument
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.2	Flow NO : Pump running against a blocked discharge (deadhead)	• V201 closed in the field by mistake (fault library) • V100 closed by air failure, ESD or LSHH-150	All motor power goes into the casing liquid (about 6 °C per minute): seal faces flash, the seal fails, hot gas oil is released; prolonged deadhead can damage the casing.	• TI-212 HI 85 °C (casing A) • TSHH-212 casing > 110 °C for 5 s → motor A tripped • FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws)	S4 L3 12 High
N4.5	Flow LESS : Weak pump pushed back in parallel	• Impeller partly blocked (fault library) • two unequal pumps in parallel at low demand (shut-off A 45.0 m vs B 46.5 m)	The weaker pump runs near shut-off: it heats, vibrates, and with a passing check valve can even be pushed backwards.	• TI-222 HI 85 °C (casing B) • TSHH-222 casing > 110 °C for 5 s → motor B tripped • II-220 HI 20 A	S3 L3 9 Medium
N5.1	Flow NO : No flow through V100	• Instrument-air failure (fault library), ESD or LSHH-150 closes V100 • V100 sticks closed	Running pump deadheaded (see N3) and the cat cracker loses its feed; D-150 empties.	• V100 fails closed (instrument air, ESD, LSHH-150) • TI-212 HI 85 °C (casing A) • TSHH-212 casing > 110 °C for 5 s → motor A tripped • FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) • LI-150 LO 30 % / HI 80 %	S4 L3 12 High
N7.1	Level MORE : D-150 fills	• Cat-cracker charge pump trips or the cat cracker cuts its feed (fault library) while the station keeps feeding • FT-100 reads low	D-150 overfills: hot oil to the cat-cracker vent or relief, liquid carry-over; if V100 is then closed, the running pump is deadheaded.	• LI-150 LO 30 % / HI 80 % • LSHH-150 at 95 % → V100 forced closed (latched until reset) • TSHH-212 casing > 110 °C for 5 s → motor A tripped	S4 L3 12 High

HAZOP recommendation

Low-flow trip of the running pump (flow in each pump leg, or FT-100 low with V100 closed, time-delayed) so a blocked discharge stops the motor before the casing gets hot.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R13

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R13
Node	N3/N4 · Pump A leg / Pump B leg	Action party	Mechanical
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.5	Composition PART OF : Strainer only partly effective	Basket torn or bypassed after cleaning	Debris reaches the impellers (fault library: impeller partly blocked): head and flow fall, vibration rises.	Strainer inspection at cleaning (procedure)	S2 L3 6 Medium
N3.3	Flow LESS : Pump delivers less than its curve	- Wear rings worn (fault library) - impeller damage - V201 disc parted from its stem (fault library)	FRC-100 opens V100 to the limit and the feed is still short; at low flow the pump runs rough and hot.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - V100 output on the faceplate	S2 L4 8 Medium

HAZOP recommendation

Pump condition monitoring: performance test against the curve every quarter, wear-ring clearances at overhaul.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R14

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R14
Node	N3/N8 · Pump A leg / Procedures	Action party	Electrical
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.4	Rotation REVERSE : Pump A turns backwards	Two supply phases swapped after electrical work (fault library)	About 60 % of the head and far less flow; the impeller nut can work loose on reverse rotation and wreck the pump.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - Commissioning checks after electrical work (procedure)	S3 L3 9 Medium
N8.11	Electrical work PART OF : Motor returned to service without a rotation check	Electrical work closed out without a bump test (fault library: phases swapped)	Pump runs backwards at the next start: low flow, possible impeller damage.	FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws)	S3 L3 9 Medium

HAZOP recommendation

Rotation (bump) test and phase-sequence check in the close-out of any electrical work on the pump motors.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R15

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R15
Node	N4/N8 · Pump B leg / Procedures	Action party	Electrical / Mechanical
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.2	Rotation REVERSE : Spare started while turning backwards	- Back-flow spins the idle pump in reverse - the operator starts it as the spare	Motor starts against reverse rotation: long high-current start, coupling or shaft overstressed, seal damage and release.	- Overload relay 49-220: > 21.7 A for 10 s → motor B tripped - Procedure: start against a closed V205	S3 L3 9 Medium
N8.7	Sequence OTHER THAN : Wrong pump or wrong valve operated	- Running pump stopped before the spare delivers - spare started against an open V205 while NRV-220 passes - wrong valve tag closed in the field (fault library: V201 closed)	Feed dip or loss; start against reverse rotation; deadhead of the running pump.	- Written procedures: start-up, change-over A → B, shutdown (Exercises tab) - FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - TSHH-212 casing > 110 °C for 5 s → motor A tripped	S3 L3 9 Medium

HAZOP recommendation

Zero-speed / reverse-rotation switch as a start permissive for each pump (or a non-reverse ratchet on the coupling).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R16

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R16
Node	N3/N4/N8 · Pump A leg / Pump B leg / Procedures	Action party	Electrical / Operations
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.1	Flow NO : No flow to the pumps	- V200 closed in error - S-200 fully blinded	Running pump starves, cavitates and runs dry: seal faces overheat and leak, gas oil fire at the pump; feed to the cat cracker lost.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - TSHH-212 casing > 110 °C for 5 s → motor A tripped - Seal-leak alarm (gas detector at each pump) - Start permissives: SIS reset, T200 ≥ 15 %, no motor trip latched, suction valve open and casing vented (both confirmed in the field)	S4 L2 8 Medium
N2.4	Composition AS WELL AS : Air or vapour drawn into the suction	- Vortex at a low T200 level - leak at the strainer cover gasket under suction	Gas pocket in the casing: loss of prime, dry running, seal failure.	- LI-200 LO 20 % / HI 85 % - LSLL-200 at 10 % for 5 s → both pumps tripped (SIS, first-out)	S3 L2 6 Medium
N3.6	Temperature MORE : Casing hot from dry running	- Casing not vented before the start (air lock) - suction valve closed - loss of prime	Seal faces run dry and fail within about a minute (model): gas oil sprays from the seal, fire.	- Start permissives: SIS reset, T200 ≥ 15 %, no motor trip latched, suction valve open and casing vented (both confirmed in the field) - TSHH-212 casing > 110 °C for 5 s → motor A tripped - Seal-leak alarm (gas detector at each pump)	S4 L2 8 Medium
N4.4	Flow NO : Spare delivers nothing when needed	- Gas pocket in the casing after draining and refilling without venting (fault library: air lock) - V206 left closed	No flow from the spare; it runs dry and the seal fails; the feed stays lost and the cat cracker trips.	- Start permissives: SIS reset, T200 ≥ 15 %, no motor trip latched, suction valve open and casing vented (both confirmed in the field) - TSHH-222 casing > 110 °C for 5 s → motor B tripped - Seal-leak alarm (gas detector at each pump)	S4 L2 8 Medium
N8.2	Sequence BEFORE : Pump started before it is vented or with the suction closed	- Venting skipped - suction valve not opened (fault library: air lock)	Dry running: the seal fails in about a minute and sprays gas oil; fire.	- Start permissives: SIS reset, T200 ≥ 15 %, no motor trip latched, suction valve open and casing vented (both confirmed in the field) - Seal-leak alarm (gas detector at each pump)	S4 L3 12 High
N8.10	Maintenance PART OF : Casing refilled but not fully vented	Vent closed when the first liquid shows, gas still trapped	Air lock on the next start: no flow, dry running, seal failure.	Start permissives: SIS reset, T200 ≥ 15 %, no motor trip latched, suction valve open and casing vented (both confirmed in the field)	S3 L3 9 Medium

HAZOP recommendation

Dry-run protection: under-power relay on each motor, and a vent sight glass to prove the casing is liquid-full before a start.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R17

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R17
Node	N3/N4 · Pump A leg / Pump B leg	Action party	Mechanical
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.8	Integrity OTHER THAN: Seal leak	Worn or damaged mechanical seal after cavitation, dry running or vibration	Gas oil at 60 °C sprays or pools under the pump; ignition on the hot motor or coupling gives a pool fire that can spread to the spare.	- Seal-leak alarm (gas detector at each pump) - ESD push-button: both pumps stopped, V100 and TV-110 closed	S4 L3 12 High
N8.9	Maintenance AS WELL AS: Casing drained to grade	Drained into an open tray or onto the ground instead of a closed drain	Gas oil spill and vapour in the pump area; fire risk.	Permit to work with isolation of the equipment (written procedure)	S3 L3 9 Medium

HAZOP recommendation

Dual mechanical seals (API 682) with seal-leak detection; seal drip pans with a kerb to the oily-water sewer.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R18

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R18
Node	N3/N4 · Pump A leg / Pump B leg	Action party	Instrument
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.9	Measurement NO : No remote view of the pump head	- PI-210 is a local gauge - II-210 is the only remote pump signal	Loss of head, wear or reverse rotation is seen late, from the flow and the valve position only.	- II-210 HI 20 A - Local gauges PI-209 / PI-210 / PI-220 on operator rounds	S2 L4 8 Medium
N4.1	Flow REVERSE : Back-flow through the idle pump	- NRV-220 flapper no longer seats (fault library) - check valve worn or held open by debris	Part of the running pump flow circulates back through the idle pump to the suction: the feed to the cat cracker is short with V100 wide open, D-150 falls, the idle pump is driven backwards.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - LI-150 LO 30 % / HI 80 % - Local gauges PI-209 / PI-210 / PI-220 on operator rounds	S3 L4 12 High

HAZOP recommendation

Casing pressure transmitters PIT-210/220 with a high alarm on the idle pump (Rev C option on the SIS tab).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R19

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R19
Node	N4/N8 · Pump B leg / Procedures	Action party	Instrument / Electrical
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.1	Flow REVERSE : Back-flow through the idle pump	• NRV-220 flapper no longer seats (fault library) • check valve worn or held open by debris	Part of the running pump flow circulates back through the idle pump to the suction: the feed to the cat cracker is short with V100 wide open, D-150 falls, the idle pump is driven backwards.	• FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) • LI-150 LO 30 % / HI 80 % • Local gauges PI-209 / PI-210 / PI-220 on operator rounds	S3 L4 12 High
N4.2	Rotation REVERSE : Spare started while turning backwards	• Back-flow spins the idle pump in reverse • the operator starts it as the spare	Motor starts against reverse rotation: long high-current start, coupling or shaft overstressed, seal damage and release.	• Overload relay 49-220: > 21.7 A for 10 s → motor B tripped • Procedure: start against a closed V205	S3 L3 9 Medium
N8.7	Sequence OTHER THAN : Wrong pump or wrong valve operated	• Running pump stopped before the spare delivers • spare started against an open V205 while NRV-220 passes • wrong valve tag closed in the field (fault library: V201 closed)	Feed dip or loss; start against reverse rotation; deadhead of the running pump.	• Written procedures: start-up, change-over A → B, shutdown (Exercises tab) • FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) • TSHH-212 casing > 110 °C for 5 s → motor A tripped	S3 L3 9 Medium

HAZOP recommendation

MOV discharge valves V201/V205 interlocked with the motors (Rev C option): open after the start, close on every stop.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R20

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R20
Node	N4 · Pump B leg	Action party	Mechanical
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.1	Flow REVERSE : Back-flow through the idle pump	• NRV-220 flapper no longer seats (fault library) • check valve worn or held open by debris	Part of the running pump flow circulates back through the idle pump to the suction: the feed to the cat cracker is short with V100 wide open, D-150 falls, the idle pump is driven backwards.	• FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) • LI-150 LO 30 % / HI 80 % • Local gauges PI-209 / PI-210 / PI-220 on operator rounds	S3 L4 12 High
N5.5	Flow REVERSE : D-150 drains back into the header	Both pumps stop (power dip, LSH-200) with V100 open and a passing check valve	Hot feed flows back from D-150 (1.5 bar g, 6 m up) through E-100, V100 and the idle pump into T200: pumps spin backwards, D-150 empties, oil above 100 °C reaches a tank that may hold water.	• Swing check valve NRV-210 / NRV-220 on each pump discharge • LALL-150 at 10 % → cat-cracker charge pump tripped (cat-cracker side)	S4 L2 8 Medium

HAZOP recommendation

Check valves: non-slam type in an inspection and seat-test programme; consider a second check valve in series on each leg.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R21

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R21
Node	N4/N5/N8 · Pump B leg / Discharge header, FE-100, V100 and FRC-100 / Procedures	Action party	Process / Mechanical
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.7	Pressure MORE : Suction side pressurised from the discharge	Back-flow through a passing check valve and an idle pump with V200 closed (suction isolated)	Suction piping, strainer and V202/V206 see the discharge pressure (up to the shut-off head); a flange or the strainer cover can leak.	- Swing check valve NRV-210 / NRV-220 on each pump discharge - Permit to work with isolation of the equipment (written procedure)	S3 L2 6 Medium
N4.3	Pressure MORE : Isolated spare casing pressurised	- V206 closed for maintenance while V205 or NRV-220 passes - liquid blocked in and warmed by the sun	Casing and suction flange (rated for suction) see header pressure or thermal expansion: leak, or a release when the casing is opened.	- Permit to work with isolation of the equipment (written procedure) - Local gauges PI-209 / PI-210 / PI-220 on operator rounds	S4 L2 8 Medium
N4.6	Temperature MORE : Idle casing heats	- Back-flow through the idle casing combined with a closed suction valve - sun on a blocked-in casing	Trapped liquid expands: flange or seal leak.	TI-222 HI 85 °C (casing B)	S2 L2 4 Low
N5.8	Pressure MORE : Blocked-in header heats up	Header blocked in between the check valves and a closed V100 (shutdown) and warmed by the sun or a fire	Thermal expansion of trapped liquid: very high pressure, flange or gasket leak.	None specific	S3 L2 6 Medium
N8.6	Sequence AFTER : Steam closed after the flow is stopped	Shutdown order reversed: V100 closed first, TV-110 later	Oil blocked in E-100 heats toward the steam temperature: coking, thermal expansion against the closed V100.	- Written procedures: start-up, change-over A → B, shutdown (Exercises tab) - TI-110 LO 140 °C (with flow) / HI 165 °C	S3 L3 9 Medium
N8.8	Maintenance OTHER THAN : Spare opened while still under pressure	- Isolation by single valves that pass - casing not drained and vented before the seal is changed	Gas oil at about 60 °C and header pressure released onto the fitter: injury, fire.	- Permit to work with isolation of the equipment (written procedure) - Local gauges PI-209 / PI-210 / PI-220 on operator rounds	S4 L3 12 High

HAZOP recommendation

Thermal relief valve on each blocked-in liquid section: pump casing / discharge between check valve and block valve, and the header between the check valves and V100.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R22

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R22
Node	N5 · Discharge header, FE-100, V100 and FRC-100	Action party	Instrument
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.5	Flow REVERSE : D-150 drains back into the header	Both pumps stop (power dip, LSSL-200) with V100 open and a passing check valve	Hot feed flows back from D-150 (1.5 bar g, 6 m up) through E-100, V100 and the idle pump into T200: pumps spin backwards, D-150 empties, oil above 100 °C reaches a tank that may hold water.	- Swing check valve NRV-210 / NRV-220 on each pump discharge - LALL-150 at 10 % → cat-cracker charge pump tripped (cat-cracker side)	S4 L2 8 Medium

HAZOP recommendation

Close V100 automatically (FRC-100 to MAN 0 %) when both pumps are stopped, so D-150 cannot drain back through a passing check valve.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R23

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R23
Node	N5 · Discharge header, FE-100, V100 and FRC-100	Action party	Operations
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.6	Flow AS WELL AS : Flow to the drain as well	Header low-point drain left open after maintenance (fault library)	Gas oil runs to the oily-water sewer (about 3 L/s): oil in the drains, fire risk in the sewer, feed short.	FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws)	S3 L3 9 Medium

HAZOP recommendation

Plug or lock the header low-point drain; keep a lock / car-seal register of drains and vents.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R24

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R24
Node	N5 · Discharge header, FE-100, V100 and FRC-100	Action party	Instrument
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.2	Flow NO : Pump running against a blocked discharge (deadhead)	- V201 closed in the field by mistake (fault library) - V100 closed by air failure, ESD or LSHH-150	All motor power goes into the casing liquid (about 6 °C per minute): seal faces flash, the seal fails, hot gas oil is released; prolonged deadhead can damage the casing.	- TI-212 HI 85 °C (casing A) - TSHH-212 casing > 110 °C for 5 s → motor A tripped - FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws)	S4 L3 12 High
N3.7	Pressure MORE : Header pressure at shut-off	Discharge blocked (V100 or V201 closed) with the pump running	The header rises to the shut-off head plus the suction head; above the piping rating only if the rating was set for the operating point.	- Shut-off head A 45 m, B 46.5 m: the pumps cannot raise the header above shut-off plus the suction head - Piping rated for the shut-off head (to confirm)	S2 L2 4 Low

HAZOP recommendation

Header pressure transmitter PIT-100 upstream of FE-100/V100 with a high alarm (blocked outlet, pump at shut-off) and a low alarm (pump lost).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R25

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R25
Node	N5 · Discharge header, FE-100, V100 and FRC-100	Action party	Instrument
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.3	Flow LESS : Pump delivers less than its curve	- Wear rings worn (fault library) - impeller damage - V201 disc parted from its stem (fault library)	FRC-100 opens V100 to the limit and the feed is still short; at low flow the pump runs rough and hot.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - V100 output on the faceplate	S2 L4 8 Medium
N5.2	Flow LESS : Feed short of the set point	- Check valve on the idle pump passing (N4) - V100 sticks while the cat cracker raises its rate (fault library) - pump wear - strainer fouling	D-150 falls; at LALL-150 the cat cracker loses its charge pump and goes to hot standby (large production loss, a hazardous restart).	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - LI-150 LO 30 % / HI 80 % - LALL-150 at 10 % → cat-cracker charge pump tripped (cat-cracker side)	S3 L3 9 Medium
N5.4	Flow MORE : Set point too high	- Remote set point from the cat cracker wrong - FRC-100 switched to a local set point in error	More feed than charge: D-150 fills; pump runs out on its curve with more power and less NPSH margin.	- FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) - LI-150 LO 30 % / HI 80 % - LSHH-150 at 95 % → V100 forced closed (latched until reset)	S2 L3 6 Medium

HAZOP recommendation

FRC-100 output-high alarm (V100 above 95 % = loop saturated) and a high clamp on the remote set point.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R26

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R26
Node	N5/N7 · Discharge header, FE-100, V100 and FRC-100 / D-150 and the cat-cracker interface	Action party	Instrument
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.3	Flow MORE : Real flow higher than measured	FT-100 reads low (fault library: impulse line / transmitter drift)	FRC-100 opens V100 and sends more than the cat cracker draws: D-150 rises; colder feed out of E-100.	- LI-150 LO 30 % / HI 80 % - LSHH-150 at 95 % → V100 forced closed (latched until reset)	S2 L3 6 Medium
N5.6	Flow AS WELL AS : Flow to the drain as well	Header low-point drain left open after maintenance (fault library)	Gas oil runs to the oily-water sewer (about 3 L/s): oil in the drains, fire risk in the sewer, feed short.	FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws)	S3 L3 9 Medium
N7.2	Level LESS : D-150 runs down	- Feed short or lost (any cause in N1-N5) - the cat cracker raises its charge (fault library)	LALL-150 trips the cat-cracker charge pump: the cat cracker goes to hot standby (large production loss, a hazardous transient).	- LI-150 LO 30 % / HI 80 % - LALL-150 at 10 % → cat-cracker charge pump tripped (cat-cracker side) - Spare pump lined up for a manual start	S3 L3 9 Medium
N7.4	Flow MORE : Charge drawn above the feed	Cat cracker raises its charge without resetting FRC-100 (fault library)	D-150 falls slowly to the low-level alarm and LALL-150.	- LI-150 LO 30 % / HI 80 % - FI-151 indication	S2 L4 8 Medium
N7.6	Control OTHER THAN : Level not controlled by the station	No level loop: D-150 is balanced only by the cat cracker resetting FRC-100	Any mismatch between the remote set point and the real charge drifts the level until an alarm.	LI-150 LO 30 % / HI 80 %	S2 L3 6 Medium

HAZOP recommendation

Flow-balance alarm FT-100 against FI-151 and the D-150 level trend, to catch a drifting FT-100 or an open drain.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R27

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R27
Node	N6 · Preheater E-100	Action party	Process
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.4	Pressure MORE : Shell at the steam supply pressure	TV-110 wide open with little condensing load (no feed flow)	The shell reaches the supply pressure (11 bar a); if its design pressure was set for the operating point, the shell is overpressured.	- TV-110 fails closed (instrument air, ESD) - Shell design pressure (to confirm)	S4 L2 8 Medium
N6.5	Integrity OTHER THAN : Tube leak or rupture	Corrosion, thermal fatigue	Steam enters the oil (water to D-150 and the cat cracker) or oil enters the shell and the condensate system.	None specific	S3 L2 6 Medium

HAZOP recommendation

Confirm the E-100 shell design pressure $\geq$ the steam supply (11 bar a) or add PSV-110 on the shell; check the tube-rupture case (API 521 two-thirds rule).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R28

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R28
Node	N6 · Preheater E-100	Action party	Instrument / Mechanical
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.5	Integrity OTHER THAN: Tube leak or rupture	Corrosion, thermal fatigue	Steam enters the oil (water to D-150 and the cat cracker) or oil enters the shell and the condensate system.	None specific	S3 L2 6 Medium
N6.6	Level MORE: Condensate builds up in the shell	Steam trap fails closed	Shell floods, duty falls; water hammer when the trap clears can break the tube bundle or a flange.	TI-110 LO 140 °C (with flow) / HI 165 °C	S3 L2 6 Medium

HAZOP recommendation

Oil-in-condensate monitor on the E-100 condensate; condensate pot or trap by-pass check.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R29

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R29
Node	N6 · Preheater E-100	Action party	Instrument
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.1	Temperature MORE : Feed outlet too hot	TV-110 driven open (TIC-110 output to 100 % or a set-point error)	Hotter feed to D-150 (light components flash); feed preheat upset for the cat cracker.	TI-110 LO 140 °C (with flow) / HI 165 °C	S2 L3 6 Medium
N6.3	Temperature LESS : Feed outlet too cold	- Steam supply pressure low (fault library) - TV-110 fails closed - E-100 fouling (fault library) - trap stuck closed and the shell floods	Cold feed to the cat cracker: lower conversion, more coke (operability).	- TI-110 LO 140 °C (with flow) / HI 165 °C - PI-111 and FI-112 indication	S2 L3 6 Medium
N6.7	Measurement OTHER THAN : TI-110 reads wrong	Thermocouple reads low (fault library)	The operator raises the steam to the false reading: real outlet too hot; with TIC-110 in AUTO the controller does it automatically.	TI-110 LO 140 °C (with flow) / HI 165 °C	S2 L3 6 Medium

HAZOP recommendation

Second outlet temperature element TE-110B with a deviation alarm, and TIC-110 in AUTO with set-point limits.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R30

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R30
Node	N7 · D-150 and the cat-cracker interface	Action party	Instrument
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.7	Flow PART OF : V100 does not close fully	Seat leakage of V100 when LSHH-150 or ESD forces it closed	D-150 keeps filling slowly after the high-high trip.	LI-150 LO 30 % / HI 80 %	S3 L2 6 Medium
N7.1	Level MORE : D-150 fills	- Cat-cracker charge pump trips or the cat cracker cuts its feed (fault library) while the station keeps feeding - FT-100 reads low	D-150 overfills: hot oil to the cat-cracker vent or relief, liquid carry-over; if V100 is then closed, the running pump is deadheaded.	- LI-150 LO 30 % / HI 80 % - LSHH-150 at 95 % → V100 forced closed (latched until reset) - TSHH-212 casing > 110 °C for 5 s → motor A tripped	S4 L3 12 High
N7.5	Measurement OTHER THAN : LI-150 reads wrong	- Level transmitter sticks or drifts - LI-150 alarm and LSHH-150 trip read one measurement	High-level alarm and trip both blind: D-150 overfills.	FI-151 vs FRC-100 seen by the operator	S4 L2 8 Medium

HAZOP recommendation

Give LSHH-150 its own transmitter and its own shut-off valve XV-150 at the D-150 inlet (today it reads LI-150 and acts through the control valve V100).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R31

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R31
Node	N7 · D-150 and the cat-cracker interface	Action party	Process
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.6	Composition AS WELL AS : Water in the feed	Water settled in the tank-farm tank or in T200 drawn with the gas oil	Water heated in E-100 flashes in D-150 (water boils at about 128 °C at 1.5 bar g): pressure surge in D-150 and steam carried to the cat cracker.	T200 water draw-off by procedure (assumed)	S4 L2 8 Medium
N7.3	Pressure MORE : D-150 overpressure	- Water or light ends flash in the hot drum - outlet blocked by the cat cracker	Drum overpressure; relief to the cat-cracker flare.	D-150 relief on the cat-cracker unit (not on PPS-PID-001)	S4 L2 8 Medium

HAZOP recommendation

Confirm with the cat-cracker unit that the D-150 relief covers hot feed carrying water or light ends.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R32

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R32
Node	N8 · Procedures	Action party	Operations / Mechanical
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.7	Pressure MORE : Suction side pressurised from the discharge	Back-flow through a passing check valve and an idle pump with V200 closed (suction isolated)	Suction piping, strainer and V202/V206 see the discharge pressure (up to the shut-off head); a flange or the strainer cover can leak.	- Swing check valve NRV-210 / NRV-220 on each pump discharge - Permit to work with isolation of the equipment (written procedure)	S3 L2 6 Medium
N4.3	Pressure MORE : Isolated spare casing pressurised	- V206 closed for maintenance while V205 or NRV-220 passes - liquid blocked in and warmed by the sun	Casing and suction flange (rated for suction) see header pressure or thermal expansion: leak, or a release when the casing is opened.	- Permit to work with isolation of the equipment (written procedure) - Local gauges PI-209 / PI-210 / PI-220 on operator rounds	S4 L2 8 Medium
N8.8	Maintenance OTHER THAN : Spare opened while still under pressure	- Isolation by single valves that pass - casing not drained and vented before the seal is changed	Gas oil at about 60 °C and header pressure released onto the fitter: injury, fire.	- Permit to work with isolation of the equipment (written procedure) - Local gauges PI-209 / PI-210 / PI-220 on operator rounds	S4 L3 12 High
N8.9	Maintenance AS WELL AS : Casing drained to grade	Drained into an open tray or onto the ground instead of a closed drain	Gas oil spill and vapour in the pump area; fire risk.	Permit to work with isolation of the equipment (written procedure)	S3 L3 9 Medium

HAZOP recommendation

Double block and bleed (or blinds) with a closed drain and vent for maintenance isolation of each pump; written isolation certificate.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R33

Project	Parallel Pump Station Ver 1 · cat-cracker feed pumps P-200A/B Operator Training Simulator	Document	PPS-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R33
Node	N8 · Procedures	Action party	Operations / Training
Drawings	PPS-PID-001 rev A (EPC sheet) and PPS-V1-PID-001 (screen P&ID), P&ID tab; PPS-PFD-100 (PFD tab)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N8.4	Timing LATE : Spare not started in time after a trip	Trip not noticed, spare not ready, operator busy elsewhere	D-150 runs down and LALL-150 trips the cat cracker.	• Motor-trip alarm in the DCS for each pump • FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) • LI-150 LO 30 % / HI 80 %	S3 L4 12 High
N8.7	Sequence OTHER THAN : Wrong pump or wrong valve operated	• Running pump stopped before the spare delivers • spare started against an open V205 while NRV-220 passes • wrong valve tag closed in the field (fault library: V201 closed)	Feed dip or loss; start against reverse rotation; deadhead of the running pump.	• Written procedures: start-up, change-over A → B, shutdown (Exercises tab) • FRC-100 LO 12.6 / HI 17 L/s (while the cat cracker draws) • TSHH-212 casing > 110 °C for 5 s → motor A tripped	S3 L3 9 Medium

HAZOP recommendation

Simulator training on the trip of the running pump, the change-over, a blocked discharge and back-flow through the idle pump; time the operator response against the D-150 hold-up.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			